Learning From Our Mistakes: Opportunities for Self-Correction Following Fast Errors Enhance Adaptive Fact Learning

Running Title: Self-Correction Boosts Adaptive Learning

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Abstract

Adaptive fact-learning systems optimize retention by tailoring review schedules to learner behaviour, yet most models consider only accuracy or the latency of correct responses, ignoring error timing. Decision-making research and dual-process theories suggest errors arise from distinct mechanisms: impulsiveness produces fast errors, whereas uncertainty yields slow errors, with corresponding differences in internal error monitoring. This implies error speed may differentially influence subsequent learning. In a spaced-repetition recognition task, we found differences in subsequent performance between fast and slow errors. Building on this, we investigated whether fast errors can be self-corrected by prompting participants to reconsider responses, using slow errors as a benchmark. Fast-latency errors were corrected significantly more often than slow-latency errors following prompting (44% vs. 24%). Accurate self-correction of fast errors improved subsequent item-level accuracy, whereas prompting slow-latency responses did not. These findings show fast and slow errors have distinct learning consequences and that latency-tailored feedback can enhance learning.

Keywords: adaptive learning, error latency, error monitoring, spaced repetition, personalized feedback, learning outcomes

1 Introduction

Adaptive fact-learning systems dynamically tailor the learning experience to each individual, using algorithms to identify domain areas that require additional attention, optimize review schedules, and maintain engagement. In contrast, traditional fact-learning methods often rely on one-method-fits-all approaches, where learners repeatedly review information without adaptation to memorization differences between individuals or materials. By personalizing content and pacing, adaptive fact-learning systems not only improve retention but also make the process more efficient and enjoyable, offering significant advantages over traditional methods. Adaptive learning systems typically work by maintaining an internal model of the learner (the "learner model"), which it continuously improves by tracking and analyzing the learner's interactions with the system (Barnes, 2005; Desmarais & Baker, 2012). Most models rely on accuracy measures (e.g., Corbett & Anderson, 1995; Klinkenberg et al., 2011), since it establishes the existence of requisite knowledge. Others focus on response latencies (e.g., Molenaar et al., 2015), as response latencies reflect the accessibility of knowledge and the efficiency of the retrieval process (e.g., Karpicke & Bauernschmidt, 2011; Pyc and Rawson, 2009). Some consider both accuracy and latency but only for correct responses, not errors, potentially ignoring important information (e.g., Collins et al., 2022; Mettler et al., 2016; van Rijn et al., 2009).

Here, we will discuss the MemoryLab model, which optimizes the sequencing of fact presentations to enhance retention and reduce forgetting. The model is an extension of the adaptive item-learning model by Pavlik and Anderson (2005; 2008) that did not incorporate information related to response latencies. In MemoryLab, response latencies on correct trials, typically about eighty to ninety percent of all trials (van der Velde et al., 2021), are used to update estimates of item activation, representing the ease with which an item could be retrieved (van Rijn et al., 2009). These updated estimates improve the model's ability to predict how easily items will be retrieved in the future, which in turn guides the scheduling of practice. The model has been shown to improve retention in experimental settings (Sense & van Rijn, 2022; van Rijn et al., 2009; Wilschut et al., 2021) and successfully applied at scale in educational settings (Sense et al., 2021; van der Velde et al., 2021) .

Currently, the response latency on the 10 to 20 percent of trials that result in an error is not used to optimize the sequencing of fact presentations. MemoryLab assumes that an error is related to a retrieval failure, for which it follows the common ACT-R assumption that a failure

occurs whenever activation drops below a fixed retrieval threshold. This insufficient activation causes the retrieval latency to exceed the time limit imposed by the retrieval threshold, ultimately resulting in retrieval failure (Lebiere et al., 1994). Because the time to register such a failure depends only on this threshold, all errors are treated as responses with the same latency. In other words, the model treats errors as uninformative about the underlying memory representation (aside from the activation being subthreshold). Yet, it is likely that an item whose activation has just dropped below the retrieval threshold would benefit from a different sequencing schedule than an item whose activation is much lower, and it is not unlikely that these items might result in different latency patterns. For instance, a near-threshold item might only need a single, well-timed repetition to stabilize its memory, while an item with much lower activation may need more intensive, repeated practice to become functional again. Moreover, one could also assume that in some cases, the random moment-to-moment fluctuations in activation could have also made the learner select an erroneous response, even though the correct response might have been retrievable given slightly more deliberation. By treating all errors as uniform, we overlook the possibility that the speed of an incorrect answer reveals how close the learner was to the correct one.

Research on the relationship between metacognition and retrieval failures clearly shows that not all retrieval failures are equivalent. A consistent finding across studies examining confidence and accuracy in memory tasks is that high-confidence errors are more likely to support later learning: when the same item is encountered again, these errors are more often corrected than those initially made with low confidence (Butler et al., 2008, 2011; Butterfield & Metcalfe, 2001, 2006; Fazio & Marsh, 2009, 2010; Kulhavy et al., 1976; Metcalfe & Finn, 2011). Although this work has primarily relied on confidence ratings, Weidemann and Kahana (2016) demonstrated that much of the diagnostic value of confidence is also captured by response times. In particular, faster errors tend to correspond to more confident responses (Kiani et al., 2014; Pleskac & Busemeyer, 2010; Rahnev et al., 2020), suggesting that retrieval-error latency carries meaningful information about the underlying state of the memory trace.

Despite the well-documented effects of post-error learning (e.g., pretesting, Mera et al., 2022), little is known about how the latency of retrieval failures specifically relates to learning outcomes, especially in fact-learning memory tasks. One promising framework for interpreting latency differences in retrieval failures comes from dual-process theories of memory.

Dual-process theories propose that recognition memory is supported by two distinct retrieval processes: familiarity and recollection. Familiarity produces a feeling that something is known; it is fast and automatic, generating a purely quantitative, strength-like signal without retrieving details from the study episode. Recollection, in contrast, involves attempts to reinstate a specific memory trace. It is a slower, more controlled process that can yield qualitative information about the original learning episode (Greve et al., 2010). Importantly, these models assume that recognition behavior typically reflects a combination of both processes, and that the relative contribution of each determines observed performance, including accuracy and response times. Evidence for this comes from work examining the time course of recognition decisions. For example, Basile and Hampton (2013) showed that visual recognition memory in monkeys unfolds across three latency-defined epochs. Very fast responses were driven largely by familiarity and were characterized by moderate accuracy and high false-alarm rates. Responses at intermediate latencies reflected contributions from both familiarity and recollection, resulting in the highest accuracy and a reduction in false alarms. This time window appears to allow initial familiarity signals to be evaluated and, when necessary, overridden by recollected information. Finally, the slowest responses were associated with guessing, yielding the lowest accuracy and an equal distribution of false alarms and misses. Together, these findings suggest that retrieval latency provides information about the underlying retrieval process engaged during an error: fast errors are more likely driven by familiarity-based false detections, whereas slow errors may reflect unsuccessful attempts to reinstate a memory trace.

Similar latency-based distinctions between errors have been developed in the decision-making literature. This research field has yielded rich insights into error processing. For example, response times on recognition memory tasks have successfully been modelled using Drift Diffusion Models to investigate the influence of misleading information on the speed of memory errors (Starns et al., 2018), and to model the relationship between confidence and error response times (Ratcliff & Starns, 2009). Thus, research in decision-making suggests that errors can differ meaningfully in their underlying processes. Insights from Drift Diffusion Models and other frameworks that conceptualize decisions as the accumulation of noisy evidence toward a threshold (e.g., Brown & Heathcote, 2008; Ratcliff 1978), and evidence from EEG research, suggest that performance errors arise via two mechanisms: Fast errors, due to impulsive actions that were not inhibited due to low motor thresholds leading to error-speeding (Novikov et al.,

2017; van Driel et al., 2012). And slow errors due to lapses in attention or high uncertainty, impairing stimulus processing and decision-making, leading to error-slowing (Böhm et al., 2014; Navarro-Cebrian et al., 2013; Novikov et al., 2017; Pailing & Segalowitz, 2004; van Driel et al., 2012; Weissman et al., 2006; Wessel et al., 2011)

Crucially, EEG research suggests that these two forms of error commission also differ in how they are subsequently detected and monitored. On the neural level, error detection is not a unitary process but relies on whether sufficient information is available for internal monitoring or whether external feedback is required (Holroyd et al., 2004; Novikov et al., 2017). Fast errors, arising from premature motor responses, typically leave enough residual stimulus–response information to support internal error detection, reflected in increased frontal midline theta activity as compared to slow errors (Cavanagh et al., 2009; Cavanagh & Frank, 2014; Novikov et al., 2017; Yeung et al., 2004). In contrast, slow errors often occur under high uncertainty, where internal cues are insufficient, making external feedback essential for detection (Holroyd et al., 2004; Novikov et al., 2017). Importantly, Novikov and colleagues (2017) demonstrated that response time profiles predict oscillatory signatures of error monitoring, emphasizing that even simple behavioral measures, such as the timing of responses, provide meaningful insight into the underlying neurobiological decision processes.

Taken together, frameworks of evidence accumulation and EEG studies converge on the idea that the speed of erroneous responses is diagnostic not only of how errors are made, but also of how they are detected and corrected, whether internally generated performance monitoring signals might already provide feedback, or whether externally provided feedback is necessary. While research in decision-making highlights the distinction between fast and slow errors, this differentiation is not applied in current adaptive learning systems. If error latency indeed signals whether an error stems from impulsiveness, or from uncertainty and/or inattention, then treating all retrieval failures as equivalent (as in MemoryLab) is theoretically questionable and limiting for model fidelity.

Extending memory models to incorporate error timing could lead to more effective, adaptive feedback. Fast errors are often detected and corrected internally (Novikov et al., 2017), suggesting that immediate presentation of the correct answer may be unnecessary in these cases. From a dual-process memory perspective, fast errors in multiple-choice tasks are also likely to reflect familiarity-based responding (Basile & Hampton, 2013), which is encouraged by the

format of the task. Rather than supplying the correct answer directly, a brief prompt for reflection ("Are you sure?") could slow responding and promote a shift toward recollection-based retrieval, supporting deeper engagement with the underlying memory representation. Whereas a slow error might warrant explicit corrective feedback. Despite this theoretical foundation, empirical work on how error timing affects learning and feedback efficacy is still largely missing.

Given the aforementioned findings, we hypothesize that fast and slow errors reflect qualitatively different underlying processes that have implications for learning. Specifically, if longer response time reflects higher uncertainty and faster responses reflect impulsive or familiarity-based responding despite intact memory representations, then fast errors should be followed by higher accuracy on later recognition than slow errors (*H1.1*). Second, following that fast errors typically occur when a trial is simple, whereas slow errors typically occur when a trial is more complex (Ratcliff and McKoon, 2008), we predict that the likelihood of fast errors will vary with item difficulty, with harder items eliciting fewer fast mistakes (*H1.2*). From a dual-process perspective, this reflects a shift from rapid familiarity-based responding for easier items toward slower, recollection-based retrieval as item difficulty increases. These hypotheses will be tested using data previously collected. To preface our results, the data showed that fast and slow errors may have distinct consequences for learning: fast errors tended to be followed by higher accuracy on later recognition than slow errors, and the ways in which errors were subsequently corrected also differed. Furthermore, the frequency of fast errors varied with item difficulty, with harder items eliciting fewer fast mistakes.

The qualitative differences observed between fast and slow errors, specifically the higher correction rates for fast mistakes and their increased frequency on easier items, suggests that response timing is a viable diagnostic for the nature of a learner's mistake. These distinct performance profiles motivated us to investigate whether latency could be leveraged to provide tailored feedback. For example, because fast errors are often detected and corrected internally (Novikov et al., 2017), feedback for these "slips" could encourage reflection (e.g., prompting "Are you sure?" and having individuals answer again) rather than providing the correct answer directly, thereby allowing the learner to resolve the error independently. Accordingly, we conducted an empirical study to assess whether fast errors are indeed detected and corrected internally more often than slow errors by prompting participants to reconsider these fast and slow responses. We hypothesized that: *H2.1*) Prompting for reconsideration would result in a higher

correction rate for fast errors compared to slow errors, given that fast errors are likely to be detected internally (Novikov et al., 2017); and *H2.2*) Including confirmation prompts after fast errors (and a subset of correct responses) would reduce overall errors and improve final accuracy relative to standard feedback.

Study 1: Retrospective Analysis

2 Methods

2.1 Participants

Data for the exploratory retrospective analysis consisted of two datasets. We utilized data collected as part of the TransplantLines study (https://umcgresearch.org/w/transplantlines; UMCG, n.d; Wilschut et al., 2025) and Klett & Balmer primary and secondary education vocabulary learning data.

2.1.1 *TransplantLines.* The TransplantLines study is a large, prospective cohort study focusing on long-term outcomes in solid organ transplant recipients and living donors, collecting extensive clinical and behavioral data. For a more detailed description of the TransplantLines Biobank and Cohort study, see Eisenga et al. (2018). The TransplantLines study protocol was approved by the local Institutional Review Board (METc 2014/077), adheres to the UMCG Biobank Regulation, and complies with the WMA Declaration of Helsinki and the Declaration of Istanbul (Eisenga et al., 2018). To assess the influence of transplant procedure on memory function, 1,470 recipients and donors participated in a MemoryLab study in which they were asked to perform three learning sessions of 8 minutes, three times self-selected without replacement from 10 unique lessons.

2.1.2 *Klett & Balmer.* The Klett & Balmer dataset consisted of English vocabulary sessions for German speaking Swiss primary and secondary education students. The dataset included 15,342 individuals, and consisted of 949 unique lessons. The individuals completed a median of 4 learning sessions per user. No demographic information was collected of the students. Data was collected by MemoryLab in accordance with its privacy policy and the Dutch Code of Conduct for Research Integrity. Analyses were conducted on anonymised data.

2.2 Model

2.2.1 *Item Scheduling Algorithm.* The scheduling of items was determined by a computational cognitive model, which extends the adaptive item-learning model proposed by

Pavlik and Anderson (2005, 2008), which has been tested in both laboratory (van Rijn et al., 2009; Sense et al., 2016; Sense et al., 2018) and educational settings (Sense et al., 2021).

Building on the ACT-R theory of declarative memory (Anderson, 2007), each item is represented as a memory chunk with an activation – the retrievability of an item – that decays over time but receives a boost whenever the item is re-encoded. A chunk (i) its activation (A) at time (t) given n previous presentations at $t_1, \ldots, t_n$ is given by the formula:

$$A_i(t) = \ln\left(\sum_{j=1}^{n} t_j^{-d_i(t_j)}\right) \tag{1}$$

When the model is queried about which item to present, the model predicts the activation of all items 15 seconds into the future. If the item with the lowest activation falls below the retrieval threshold, it is scheduled for presentation. If all predicted activations remain above the threshold, a new item is introduced. By selecting items based on activation levels, the model maximizes spacing (for a review, see Dempster, 1988) between repetitions while ensuring that most responses are correct.

Parameter d from Equation 1 represents the decay of activation. The higher the decay, the faster a chunk's activation decreases, leading to it being repeated sooner than an item with a lower decay. To account for differences in item difficulty on decay, an offset labeled as *speed of forgetting* ($SoF;\ a_i$) is included. The decay (d) of a chunk (i) at a given time (t) depends on the activation of the chunk at its previous encounter, as well as the item difficulty offset (a_i).

$$d_i(t) = c * e^{A_i(t_{n-1})} + a_i \tag{2}$$

Each item has a standard initial SoF when first presented, which is updated during learning. At each presentation, the model calculates an expected response time, $E(RT)$, based on the activation, by taking the negative inverse (e^{-A_i}; Anderson et al., 1998). Additionally, the estimated reading time of the prompt (number of characters) is taken into account (Nijboer, 2011). The estimated reading time is set to 300ms for figure prompts.

$$E(RT) = e^{-A_i} + reading\ time \tag{3}$$

The model updates the SoF parameter based on response accuracy and, for accurate responses only, on the mismatch between expected and observed response time. A slower than expected correct response suggests greater memory decay than estimated, indicating a higher forgetting rate. An incorrect response implies the item's activation dropped below the retrieval threshold, whereas it was expected to be above the threshold, warranting an increased forgetting rate. Conversely, a faster than expected correct response suggests a stronger memory trace, leading to a lower forgetting rate. To prevent extreme response times (e.g., due to distraction or a small break) from distorting estimates, observed times are capped at 1.5 times the expected retrieval time of an item with an activation at the level of the retrieval threshold. The model then refines the forgetting rate using a binary search within a small window around the previous value to find the speed of forgetting that minimizes the mismatch between predicted and observed response times.

2.2.2 *Fast/Slow Errors.* To investigate the effect of error response times on accuracy and response times at a subsequent trial of the same item, an approach to quantify fast versus slow errors is needed. In the context of decision-making, fast errors are typically even faster than correct responses, whereas slow errors are typically slower than correct responses (Ratcliff et al., 2016). Taking this into account, since we already have a way to estimate the expected reaction time of a correct response (Equation 3), we decided to quantify fast errors as errors with a quicker observed response time than a correct response would be estimated to have (Figure 1B). Flowing from that, slow errors were quantified as errors with a slower observed response time than estimated by the activation of the chunk (Figure 1C).

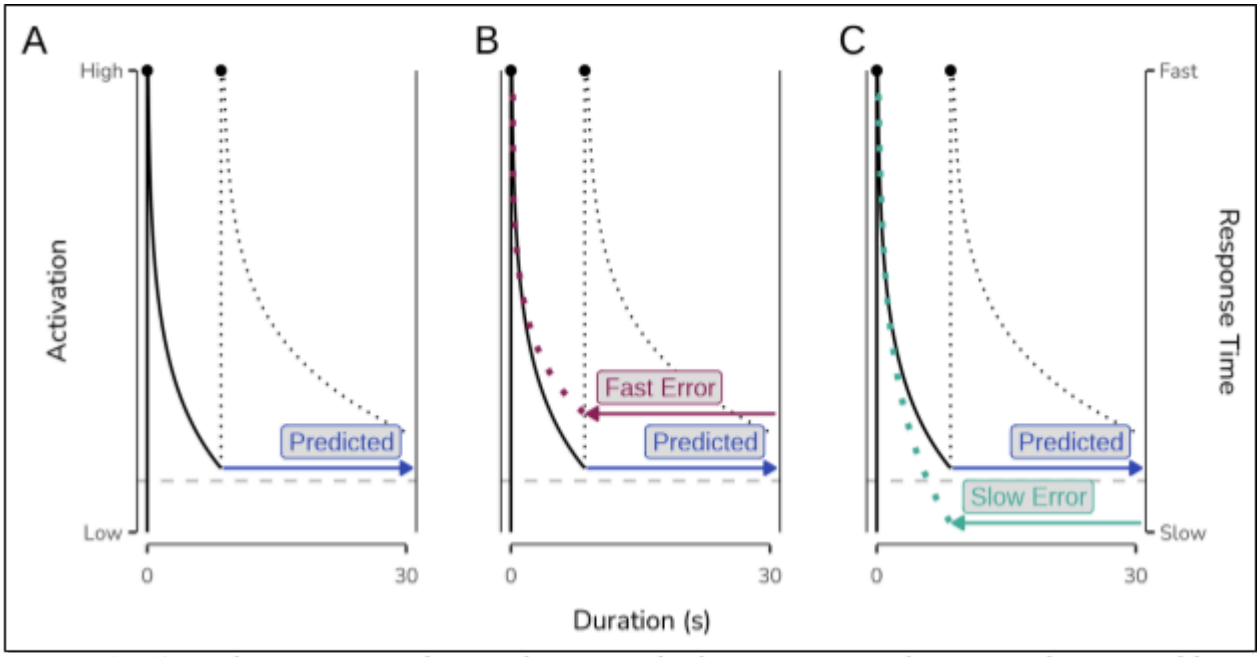

Figure 1. A) Each presentation boosts the item to high activation, making it easily retrievable and fast to answer. Activation then decays rapidly. The model quizzes the item just before it is likely to be forgotten and predicts the response time that would occur if the item is successfully retrieved. Based on this, an error can be either: B) faster than the predicted response latency (a fast-latency error); or C) slower than the predicted response latency (a slow-latency error).

2.3 Analysis

Several steps were taken to ensure data quality for both data sets. First, if participants completed the same lesson multiple times, we only included their first session. Second, sessions were removed if a break of 60 seconds or more occurred without a response, as this indicated an interruption in the task. Third, sessions that were either too short or too long were excluded; only sessions lasting between 2 and 15 minutes and containing at least 25 trials were retained. Finally, to ensure meaningful data, sessions with an average accuracy below 0.5 or a median reaction time exceeding 8 seconds were excluded. To investigate how feedback type and prior response behavior influence learning outcomes, we conducted descriptive summaries and visualizations as well as two mixed-effects models. Trial accuracy (coded 1 = correct, 0 = incorrect) was regressed on *user age* (in years, centered; only for TransplantLines data), *lesson* (categorical, with the first lesson as the reference level), and *response type at the previous presentation of the same item* (categorical: correct [reference level], fast error, slow error) using a logistic mixed-effects regression (GLMER) with random intercepts for user and item. Log-transformed response times

were regressed on the same predictors using a linear mixed-effects regression (LMER) with random intercepts for user and item. For the Klett & Balmer data, we only kept lessons with more than 100 unique sessions, to reduce the number of slopes that had to be estimated. The analysis was carried out using the software language R, version 4.4.0 (R Core Team, 2024), and the software R-Studio, version 2024.04.0 (Posit Team, 2024). The packages used for performing the GLMER were lme4 (Bates et al., 2015), car (Fox & Weisberg, 2019), ggeffects (Lüdecke, 2018), DHARMa (Hartig, 2016).

2.4 Procedure

2.4.1 *TransplantLines.* Participants were invited via email to complete three online MemoryLab sessions. Each session consisted of image-term questions with four multiple-choice answer options. Items were introduced adaptively using the adaptive testing algorithm designed for optimal spacing as described in 2.2.1. Invitations for subsequent sessions were sent one week after participants completed the previous session. For each session, participants could choose one topic from a set of ten available topics.

2.4.2 *Klett & Balmer.* Lessons were offered as part of the school curriculum, but students retained the option to revisit them independently outside the assigned activities. Items were introduced adaptively using the adaptive testing algorithm designed for optimal spacing as described in 2.2.1.

3 Results

3.1 Data Cleaning

3.1.1 *TransplantLines.* A total of 312 sessions were omitted due to completion of the same lesson multiple times. In these cases we only kept the first session. Another 10 sessions were removed from the data for containing a long interruption of at least 60 seconds. No sessions exceeded the duration limits. Finally, 6 sessions were dropped for having an average accuracy below 0.5 or a median reaction time exceeding 8 seconds, leaving a total of 3172 learning sessions.

3.1.2 *Klett & Balmer.* After checking for duplicates, 56541 sessions were removed. We removed 8492 sessions which had an interruption of at least 60 seconds. Too short (i.e., below 2 minutes) and too long (i.e., above 15 minutes) session duration meant that 33793 more sessions were dropped for the analysis. Finally, 214 sessions were dropped for having an average accuracy

below 0.5 or a median reaction time exceeding 8 seconds. A total of 42251 sessions remained. For fitting the (G)LME, another 16448 sessions were excluded as they were part of the lessons that did not have more than 100 unique sessions.

3.2 Behavioral Descriptives

***3.2.1** TransplantLines.* The distribution of response time and mean accuracy is visualized in Figure 1 and Figure 2 respectively, across all participants and grouped by age bins. The response time had a weighted mean of 3.81s (*SD* = 1.21), whereas weighted mean accuracy was 0.86 (*SD* = 0.12).

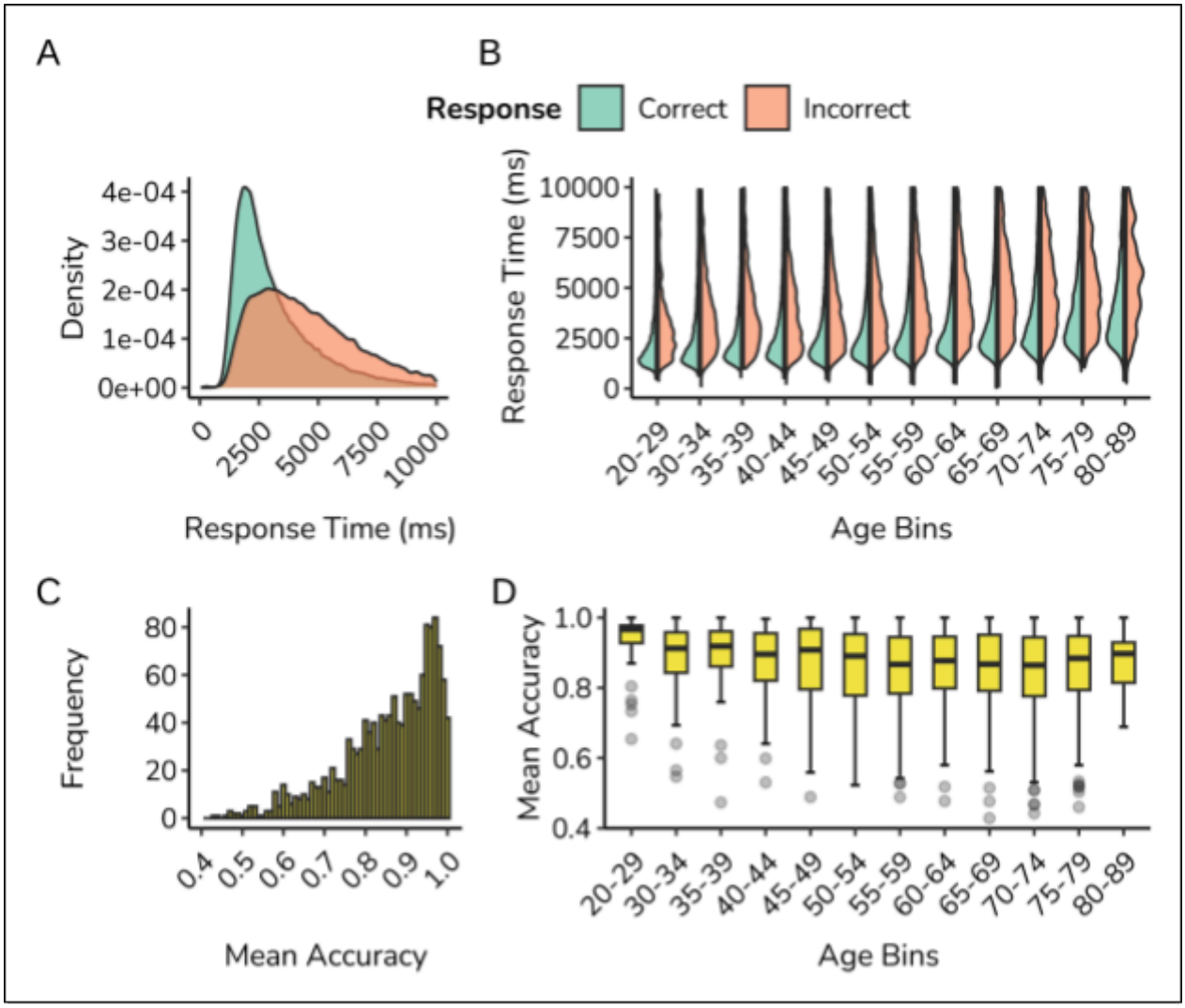

Figure 2. TransplantLines: Distribution of response times and user accuracy for all participants (A;C) and grouped by age bins (B;D)

3.2.2 *Klett & Balmer.* A visualization of the distribution of response time and mean accuracy can be found in Figure 3. The response time controlled for each individual had a mean of 3.68s (*SD* = 1.17), whereas mean accuracy was 0.94 (*SD* = 0.07).

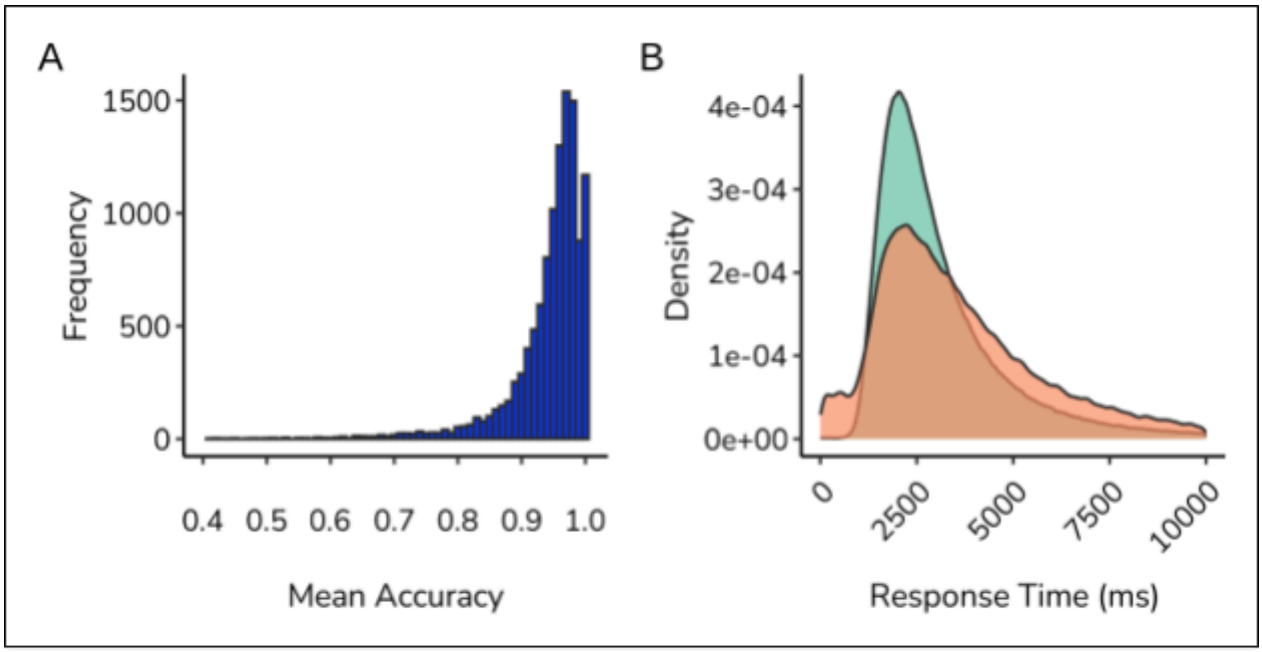

Figure 3. Klett & Balmer: Distribution of mean accuracy (A) and response times (B)

3.3 Error Type and Subsequent Trial Behavior

A visualization of the distribution of *accuracy at the subsequent trial of the same item* following a specific error of commission can be found in Figure 4 for both datasets. Visual inspection suggests that, across all individuals and within most age bins, accuracy at a subsequent trial of the same item seems to be higher if the initial error was associated with a short response time.

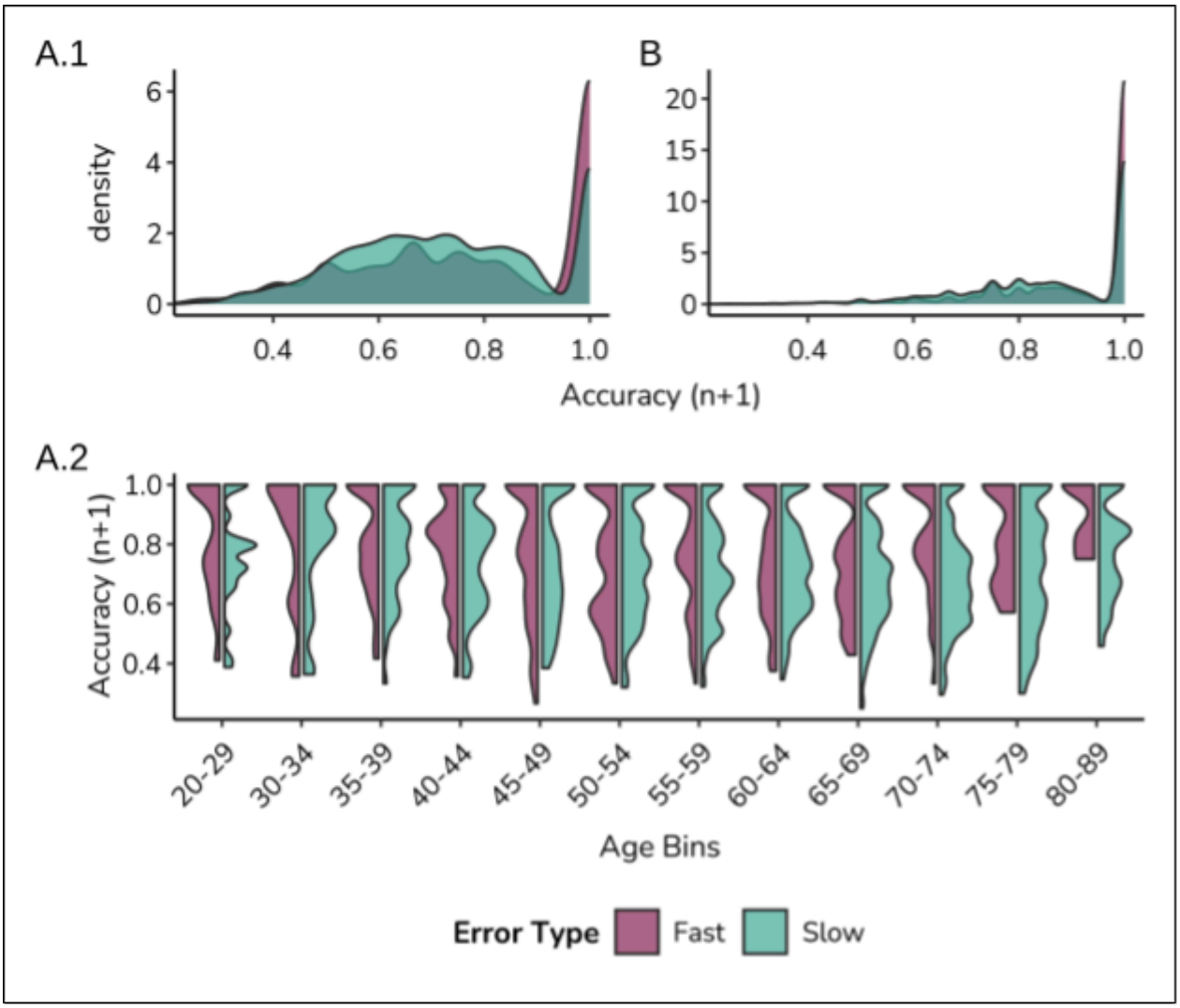

Figure 4. TransplantLines: Distribution of accuracy at the subsequent trial of the same item (n+1) by error type for all participants (A.1) and grouped by age bins (A.2). KuB: Distribution of accuracy at the subsequent trial by error type for all participants (B)

3.3.1 *TransplantLines.* The ANOVA tables of the LMER of response time and GLMER of correctness can be found in Table 1 and Table 2 respectively. Additionally, a plot of the estimated marginal effects of response type at the previous trial of the same item can be found in Figure 5. From here, it can be observed that there is a clear effect of the error type on both the response time and the accuracy on a subsequent trial of the same item. Response times were slower following errors, particularly slow errors, and accuracy likewise dropped most when the preceding error had been slow.

Table 1: Response Time (n+1; TransplantLines) ANOVA table

Predictor	Sum of Squares	*DF*	Mean Square	*F*	*p*
Age	162.46	1	162.46	792.95	<.001*
Lesson	1843.14	9	204.79	996.07	<.001*
Response Type	1204.91	2	602.46	2022.38	<.001*

Analysis of Variance (Type II, Satterthwaite) table of the LME of log transformed response times on the subsequent trial of the same item.

Table 2: Accuracy (n+1; TransplantLines) ANOVA table

Predictor	X^2	*DF*	*p*
Age	57.12	1	<.001*
Lesson	4781.54	9	<.001*
Response Type	3746.23	2	<.001*

TransplantLines: Analysis of Deviance (Type II, Wald) table of the GLME of accuracy on the subsequent trial of the same item.

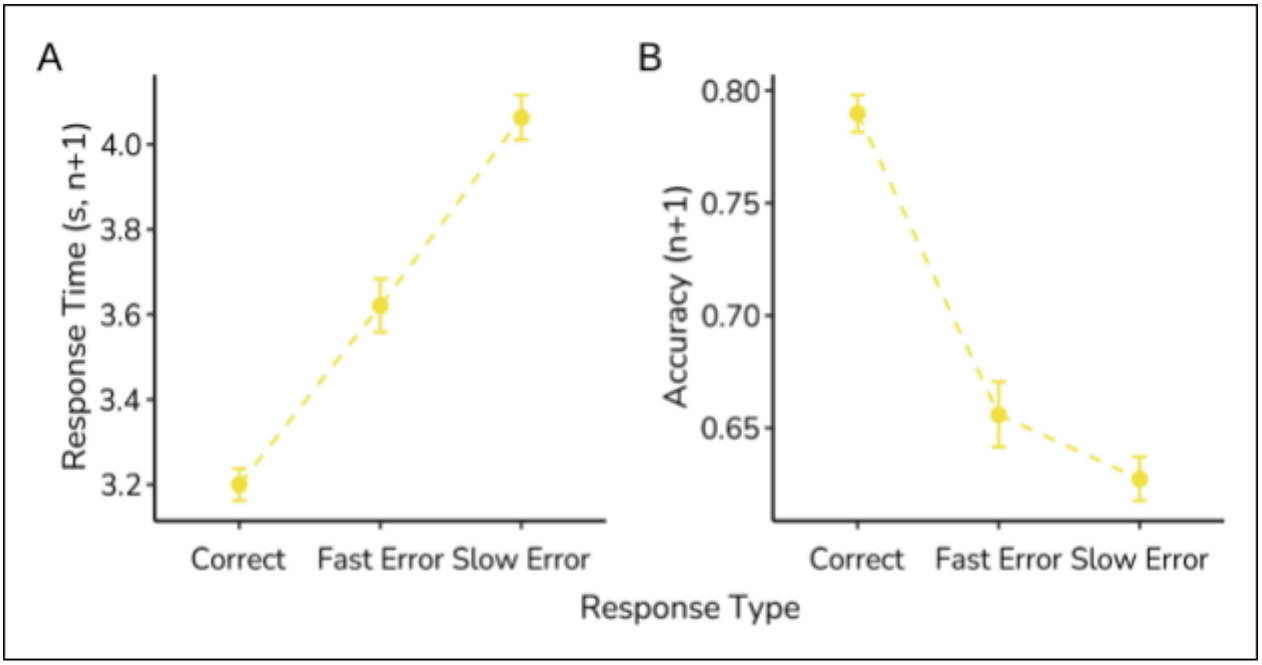

Figure 5. TransplantLines: Estimated marginal effects of error type on response time (A; log-inversed) and accuracy (B) at subsequent same-item trial (n+1) . Error bars represent the 95% confidence intervals.

3.3.2 *Klett & Balmer.* The ANOVA tables of the LMER of response time and GLMER of correctness can be found in Table 3 and Table 4 respectively. A plot of the estimated marginal effects can be found in Figure 6. Similar effects as the ones found for the TransplantLines data can be found in the Klett & Balmer dataset: response time increases and accuracy decreases when the previous response was an error, especially when that error was slow.

Table 3: Response Time (n+1; Klett & Balmer) ANOVA table

Predictor	Sum of Squares	*DF*	Mean Square	*F*	*p*
Lesson	12903.20	117	110.28	405.3	<.001*
Response Type	2031.28	2	1015.64	3732.5	<.001*

Klett & Balmer: Analysis of Variance (Type II, Satterthwaite) table of the LME of log transformed response times on the subsequent trial of the same item (n+1).

Table 4: Accuracy (n+1; Klett & Balmer) ANOVA table			
Predictor	X^2	*DF*	*p*
Lesson	8263.30	117	<.001*
Response Type	5895.50	2	<.001*

Klett & Balmer: Analysis of Deviance (Type II, Wald) table of the GLME of accuracy on the subsequent trial of the same item (n+1).

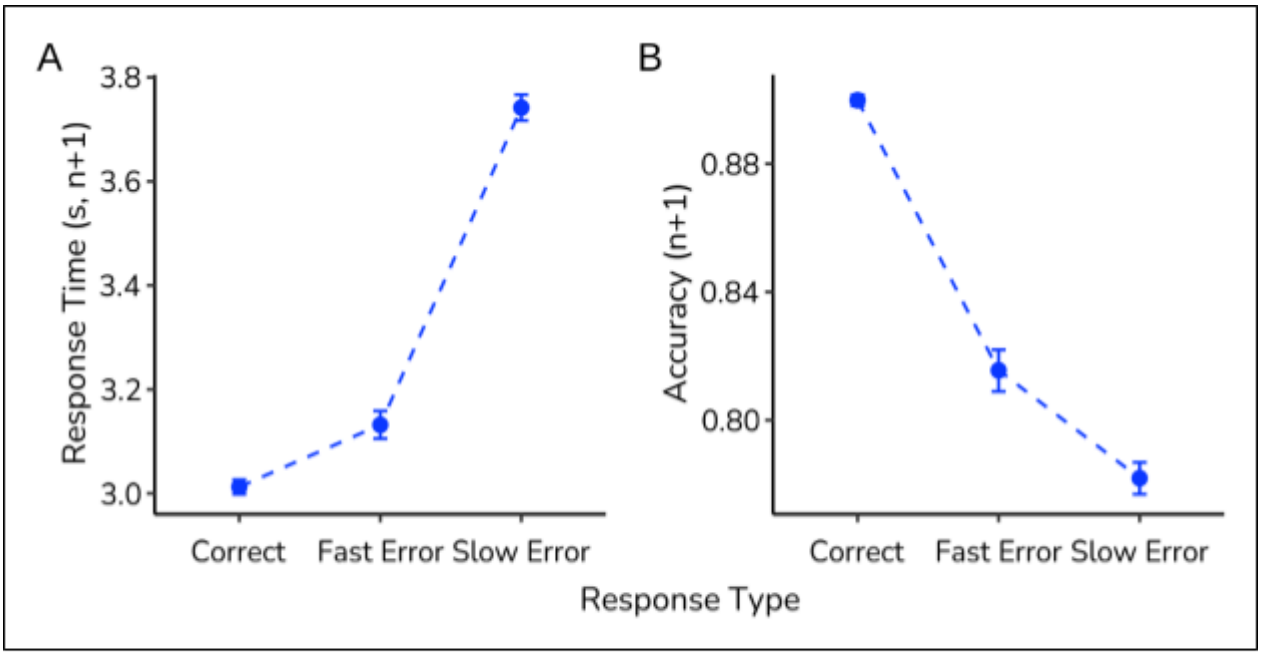

Figure 6. Klett & Balmer: Estimated marginal effects of error type on response time (A; log-inversed) and accuracy (B) at subsequent same-item trial (n+1). Error bars represent the 95% confidence intervals.

3.4 Error Type and Duplicate Errors

The relationship between error types and the repetition of specific item–response pairs is visualized in the alluvial plot in Figure 7. This figure displays repetition sequences in which other item–response pairs for the same item were allowed to occur between repeated errors (i.e., *interrupted* sequences). Across both datasets - a consistent pattern emerges: slow errors are more likely to result in the same incorrect response than fast errors, and when they are repeated, they are also more likely to be slow again. For instance, in the TransplantLines data, 84.4% of

repeated errors were slow on the first occurrence, and of those, 84.0% were slow again on the second occurrence. Of the 15.6% of repeated errors that were fast on the first occurrence, only 63.9% were slow on the second occurrence. This asymmetry in repetition patterns (TransplantLines, X^2 (1) = 157.89, $p < .001$; Klett & Balmer, X^2 (1) = 479.83, $p < .001$) suggests a qualitative difference in how fast and slow errors behave over time.

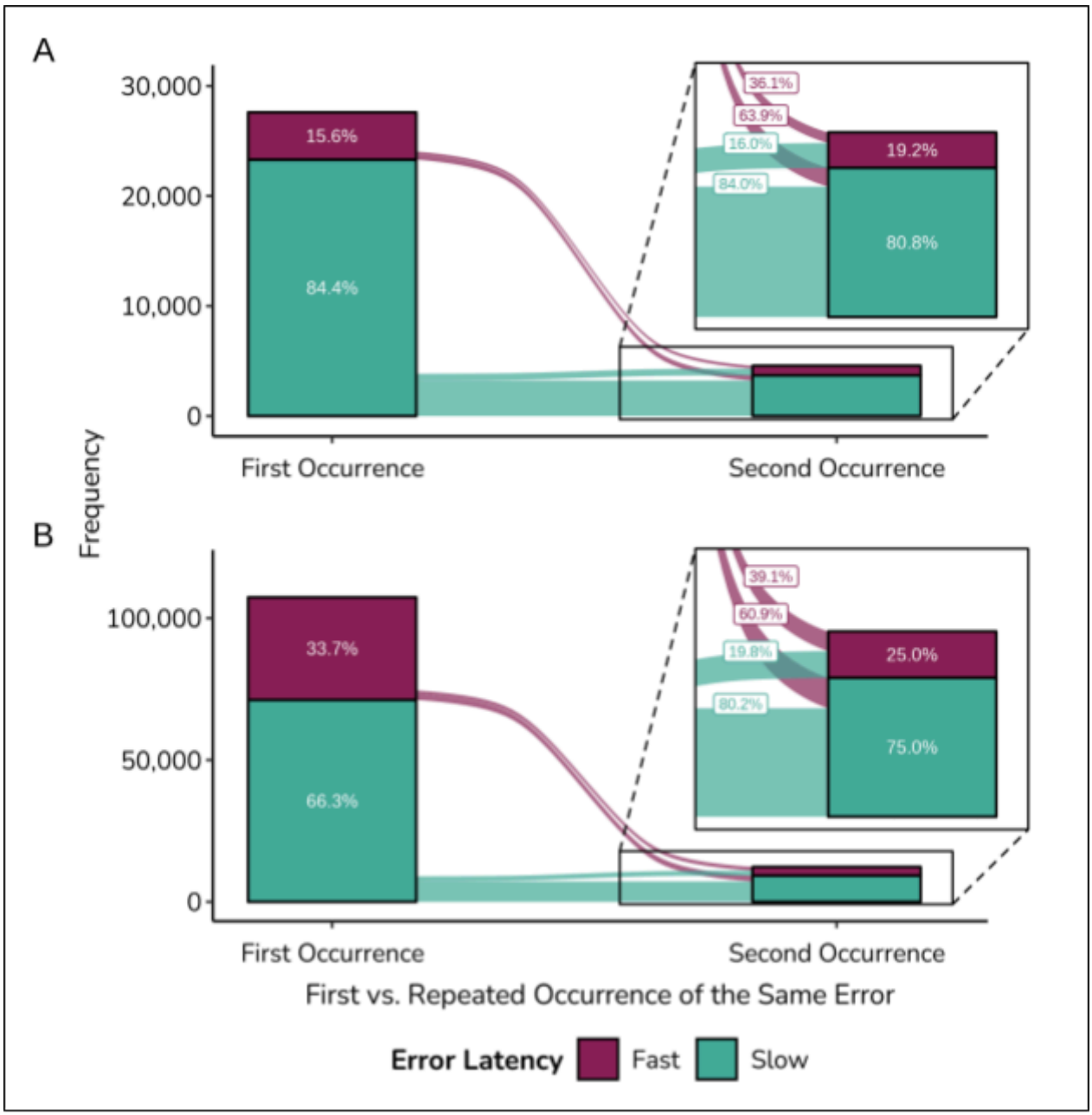

Figure 7. Alluvial plot showing error repetition patterns by error type. In this analysis, repeated errors are counted even when other item–response combinations occur between them. For example, the sequence Italy–Paris, Italy–Rome, Italy–Paris, Italy–Paris counts as two repeated errors. Panel A shows data from the TransplantLines dataset; Panel B shows data from the Klett & Balmer dataset.

3.5 Error Type and Item Difficulty

The relationship between error type and item difficulty was plotted as a scatterplot between the proportion of fast errors out of all errors, and mean accuracy on an item, which can be found in Figure 8. It can be observed that there is a positive linear relationship between the proportion of fast errors and the accuracy on an item, suggesting that fast errors become more prevalent as items become easier.

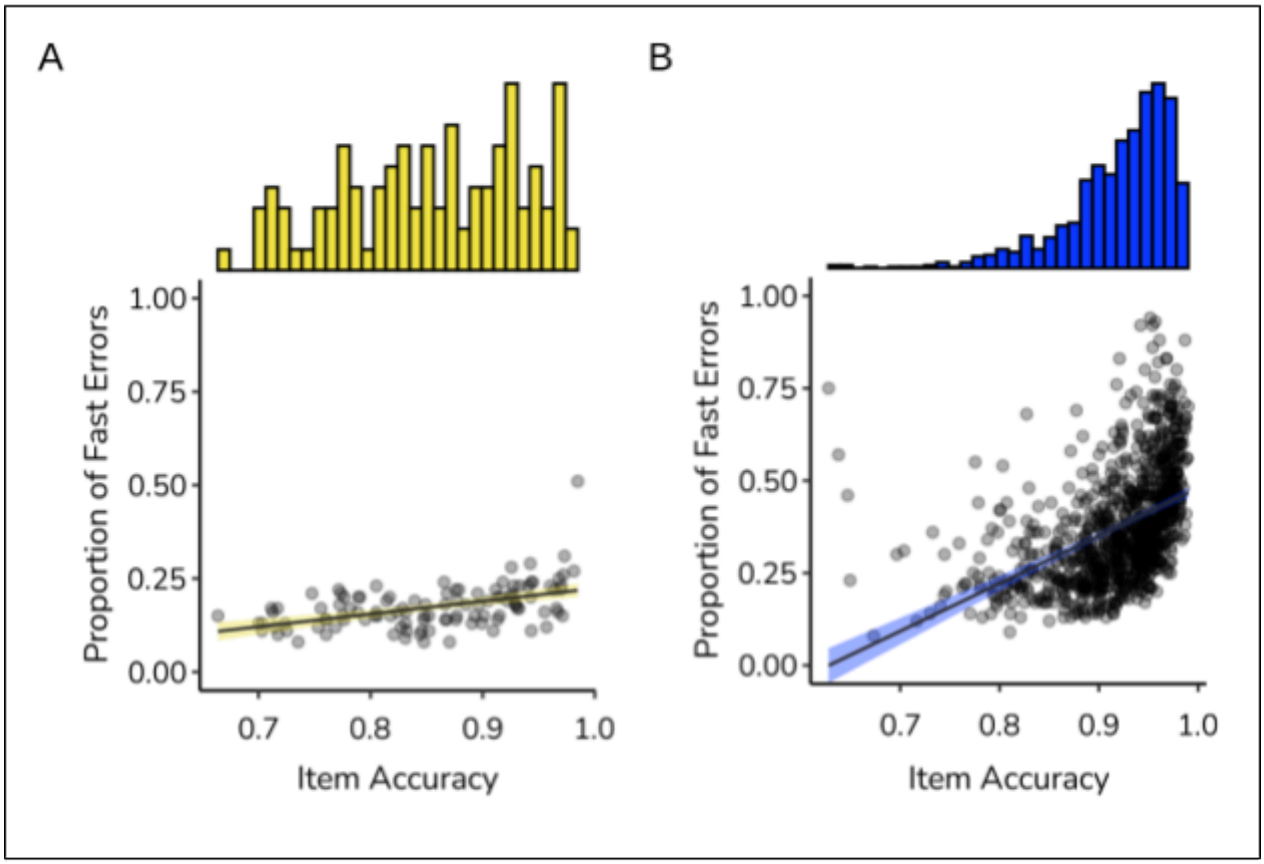

Figure 8. Relationship between item accuracy and proportion of fast errors, shown separately for TransplantLines (panel A) and Klett & Balmer (panel B). The marginal distribution of item accuracy is displayed above.

4 Intermediate Discussion

To investigate how error response times influence subsequent recognition memory performance, we analyzed historical MemoryLab recognition learning task data, consisting of data from the TransplantLines study and data obtained from primary and secondary education students using Klett & Balmer educational materials. We classified each error as fast or slow depending on whether its observed response time fell below or above the model-estimated response time. On average, fast errors were followed by 4% higher accuracy on subsequent trials of the same item compared to slow errors, consistent with our hypothesis.

The data also revealed additional distinctions between fast and slow errors. When the same incorrect answer–response pairing recurred, slow errors tended to recur as slow errors approximately 80% of the time, whereas fast errors recurred as fast errors only about 40% of the time. This pattern suggests that slow errors may reflect more stable or persistent misconceptions. Furthermore, we observed that fast errors became more likely as item difficulty decreased, indicating a tendency toward impulsive or incidental mistakes when the retrieval process is likely to be easy, which is in line with the work of Ratcliff and McKoon (2008).

Together, these results indicate that variations in error response latency may signal differences in the processes contributing to recognition memory errors. Distinguishing between fast and slow errors - paralleling distinctions commonly made in decision-making research - could be used to optimize learning.

<u>Study 2: Experimental Feedback Intervention</u>

Our analysis of previously collected spaced-repetition retrieval practice with multiple-choice questions data revealed that fast errors result in higher accuracy on subsequent trials of the same item compared to slow errors. The incongruencies are likely related to the underlying processes: where fast errors are often linked to impulsive responding, slow errors are more likely to reflect insufficient encoding or difficulty retrieving the item. Building on these distinctions, and findings that fast errors are more likely to be detected internally as compared to slow errors, we investigated whether fast errors can be self-corrected by prompting participants to reconsider responses, using slow errors as a benchmark.

5 Methods

5.1 Participants

The participant sample consisted of $N = 77$ first-year psychology students from the University of Groningen. Participants were recruited through a first-year psychology student participant pool, where students take part in research in exchange for study credits. Eligibility criteria included normal or corrected-to-normal vision and an age range of 18-25 years. Participants were encouraged to have a full night's sleep prior to the experiment and were asked to reschedule if they felt they had slept poorly. No other demographic information was collected from the participants. On the basis of a checklist developed by the EC-BSS at the University of Groningen, the study was granted exemption from full ethics and privacy review by the EC-BSS.

5.2 Model

5.2.1 *Item Scheduling Algorithm.* The scheduling of items was determined by the algorithm described in 2.2.1.

5.2.2 *Error Quantification.* Based on the results from the exploratory analysis we decided to divide the errors in *fast-latency*, *mid-latency*, and *slow-latency* errors, retaining the previous definition of a fast-latency error (see Figure 1B) but splitting slow-latency errors (Figure 1C) into two categories. Slower than fast-latency responses were split into mid-latency and slow-latency errors to specifically target the very slow responses, and not interrupt the flow of the learning task too much. Errors are defined as incorrectly chosen answer options. As before, fast-latency errors are errors with a quicker than estimated observed response time. We defined slow-latency errors as those with an observed response time that exceeded 1.5 times the time needed to retrieve a chunk at the model's retrieval-threshold (3788 ms). The idea is that, under the model, incorrect responses are generally expected to cluster around the retrieval threshold. By adding a margin (here 1.5×), we restrict intervention to the genuinely slow errors, cases that likely reflect processes beyond normal retrieval difficulty (the effect of this quantification on accuracy at a subsequent trial of the same item is displayed in Figure S1 in the supplements). This cutoff is somewhat arbitrary, and identifying optimal intervention boundaries remains an open question for future work. Flowing from that, mid-latency errors were quantified as errors with a slower observed response time than estimated from the activation of the chunk, but faster than slow-latency errors as described above.

5.2.3 *Feedback Manipulation.* During the manipulation blocks, the feedback was tailored to the error type. The manipulation involved a confirmation-prompt feedback mechanism that was implemented in two distinct experimental blocks: one targeting fast-latency errors and the

other targeting slow-latency errors. The prompt offered participants the opportunity to confirm or revise their initial response. To avoid participants learning that the confirmation prompt signaled an error (i.e., treating the prompt as an error cue), the prompt was also introduced during a subset of correct trials with with latencies aligned to the block's manipulation (e.g., a *correct* fast-latency response during a fast-latency block).

Considering that the prompt increases the time spent on one item, it should only be used when that extra time investment is worthwhile. Based on the theoretical framework described above, we assume that intervening on errors is most productive (in terms of correction rate) when learners are aware that they made an error. Learners are most likely to be aware of their own errors when those errors are fast. To avoid disrupting the flow of the experiment, mid-latency errors were therefore not intervened upon. The specific implementation details of the confirmation prompt and the algorithm used for ambiguity maintenance are described in Section 5.4.

5.3 Stimuli

The stimuli consisted of country outlines obtained from the *rnaturalearth* package (https://github.com/ropensci/rnaturalearth). We created three stimulus sets corresponding to Africa, Asia, and Europe plus Latin America. For each set, we selected the 25 most difficult countries for that continent based on prior knowledge norms of country outlines reported by Wilschut and colleagues (2023). Because these norms were derived from open recall rather than recognition, several of the selected outlines turned out to be overly distinctive in a pilot version of our task, leading to ceiling performance and rendering them uninformative. We therefore manually replaced these items with other countries from the same continent for which no prior knowledge norms were available. In total, 22 outlines were replaced. We also removed several islands belonging to the outlines of Ecuador, Yemen, and South Korea, as their inclusion altered the aspect ratio of the images making them more distinctive. Finally, we filled in the Baikonur Cosmodrome indentation in the outline of Kazakhstan to avoid an atypical contour shape. An overview of all stimulus modifications is provided in Table S1 the Supplementary Materials.

Distractors were selected on a trial-to-trial basis. On each trial, three distractors were semi-randomly sampled from the same stimulus set. For one of the three distractors, sampling followed a hierarchical constraint: when possible, the distractor was selected from countries belonging to the same United Nations subregion; if no other stimuli in the set belonged to the

same subregion, the distractor was instead sampled from countries belonging to the same World Bank region. After applying this subregion/region constraint, the distractor was preferentially selected from countries that had already been studied, when possible. The remaining two distractors were sampled from already studied countries whenever possible; if insufficient studied countries were available, they were sampled from the full stimulus set.

5.4 Experimental Setup

The experiment was developed and run using OpenSesame (4.1.3; Mathôt et al, 2012) and custom Python (3.11.3) scripts, which allowed for precise control over stimulus presentation, response timing, and feedback logic. The adaptive scheduling of items and feedback manipulation were implemented through custom code, integrating the MemoryLab model's activation-based scheduling framework. Response times were recorded in milliseconds from stimulus onset to the initial keypress. Stimuli were presented at a fixed resolution and size (700x700 pixels) to ensure consistency across trials. The experimental interface was presented full-screen on a standard 24-inch monitor at a resolution of 1920×1080 pixels. Input was collected via a standard USB keyboard.

Each trial began with the immediate presentation of a stimulus, which remained on screen until the participant responded using the number keys (1–4), or the spacebar in case of a new item. Upon response, feedback was presented without delay. For standard correct responses, feedback was shown for 1000 milliseconds, after which the task automatically progressed to the next trial, following a 200ms inter-trial interval (ITI). For erroneous standard feedback trials, the 1000ms feedback was followed by a prompt instructing the participant to press the spacebar to continue, after which the next trial followed after a 200ms ITI.

During the control block/condition, only the standard feedback procedure was utilized. For the two other experimental blocks, the standard feedback procedure was replaced with a confirmation-prompt feedback mechanism. The condition for triggering this mechanism depended on the block: one block implemented the confirmation prompt only for fast errors, and the other only for slow errors. In these instances, participants who committed an error with the type corresponding to the block were immediately presented with an "Are you sure?" prompt and were instructed to select an answer for a second time (for a visualization, see Figure 9). This process offered the opportunity to confirm or revise their initial response. To prevent participants from treating the confirmation prompt as an error cue, the prompt was also introduced during a

subset of correct trials with a latency corresponding to the blocks' manipulation (e.g., a *correct* fast-latency response during a fast-latency block). The frequency of prompting during correct trials was matched to the rate of error-prompting within that respective block's speed category (fast or slow) using the following equation:

$$Prompt\ Probability = \frac{(Ep + 1) - Cp}{6} \quad (4)$$

Where *Ep* is the number of error prompts, and *Cp* the number of correct prompts. Here, the probability was constrained between 0 and 1. In short, the probability of a confirmation prompt happening at the current correct trial with the respective block's speed manipulation depended on the mismatch between the number of error prompts and the number of correct prompts. We adopted '*Ep* + 1' to allow prompts to appear even before the first error had occurred, thereby maintaining a non-zero probability of a correct latency-matching prompt from the start of the task. The denominator sets the baseline probability of receiving a prompt on correct trials when *Ep*=*Cp*, of 1 in 6 (16.7%). It was introduced to ensure that an error-prompted trial was not always followed by a prompt on the next correct trial (i.e., without it, participants could quickly learn that every error-prompted trial is followed by a correct-prompted trial). The second response time and accuracy were then registered, where the response time of the second response was measured from the moment the prompt appeared on the screen. The accuracy and response time of the second response were not used to inform the model, for that the first response was used. Following the registration of the second response, the trial concluded the same way as standard feedback trials. Responses that did not meet the speed criterion for the specific block (e.g., a fast response occurring in the block designated for slow errors) were handled identically to responses in the standard feedback condition.

Figure 9. Illustration of a confirmation-prompt trial.

5.5 Procedure

Upon arrival at the research laboratory, participants were provided with an information sheet outlining the study's purpose and procedures and were asked to provide written informed consent. The entire session lasted approximately 35 minutes.

After consent, participants received instructions indicating that responses should be made using the number keys 1 to 4, and that the spacebar should be used to continue through the task. They were informed that the experiment consisted of three blocks and for each block they were instructed whether they would receive prompts for errors, but they were not instructed about the speed category. The experiment started off with a 2 minute practice round, and an explicit demonstration of how the confirmation prompt worked. Each experimental block had a duration of 10 minutes.

The experiment followed 1 of 36 (3!✕3!) randomly shuffled counterbalanced versions, assigned based on participant number to ensure an even distribution across conditions. These versions varied along two dimensions: first, the order of the manipulation (control, fast-latency, slow-latency); and second, the order in which stimulus sets were learned (Africa, Asia, Europe/Latin America). This counterbalancing controlled for potential order and stimulus set effects.

5.6 Analysis

To investigate how feedback type and prior response behavior influence learning outcomes, we conducted a single generalized linear mixed-effects regression (GLMER) analysis, supplemented by descriptive summaries and visualizations. The analysis was carried out using the software language R, version 4.4.0 (R Core Team, 2024), and the software R-Studio, version

2024.04.0 (Posit Team, 2024). The packages used for performing the GLMER are lme4 (Bates et al., 2015), car (Fox & Weisberg, 2019), ggeffects (Lüdecke, 2018), DHARMa (Hartig, 2016).

Prior to analysis, the data were visually inspected for aberrant patterns in response accuracy and timing. Here, we applied the same criteria as in the retrospective data analysis.. Moreover, sessions were checked on whether performance dropped significantly over time, as indicated by observable differences in response times and accuracy. Descriptive statistics and visualizations were used to explore accuracy and response time distributions across conditions. This included summaries by feedback condition intended to inform interpretation of the model-based results.

To assess the effect of the confirmation prompt, the primary model was a GLMER predicting trial-level accuracy (correct, incorrect) of the confirmed/revised response (i.e., the registered response after the confirmation prompt) solely for prompt trials, with random intercepts for both participants and items/facts to account for repeated measurements. The fixed effects included block number (#1, #2, #3), the stimulus set (Africa, Asia, Europe & Latin America), the latency type (slow-latency, fast-latency), the initial correctness (i.e., the registered response before the confirmation prompt: incorrect, correct), and the interaction between the latency type and the initial correctness. All fixed effects were automatically dummy coded (0-1) by the lme4 package, with the first-mentioned category as specified above as the reference. Assumptions were checked visually by means of residual plots. Statistical significance of predictors was evaluated using Type II Wald chi-square tests. Post-hoc comparisons of the interaction effect were conducted to assess differences in final accuracy between latency-types within initially incorrect, and within initially correct responses.

The second model was a GLMER predicting trial-level accuracy (correct, incorrect) of the final response (i.e., the initial response for non-prompted trials and the confirmed/revised response for prompted-trials) for all trials, with random intercepts for both participants and items/facts to account for repeated measurements. The fixed effects included block number (#1, #2, #3), the stimuli set (Africa, Asia, Europe & Latin America), and the experimental condition (control, fast-latency, slow-latency). All fixed effects were automatically dummy coded (0-1) by the lme4 package, with the first-mentioned category as specified above as the reference.

6 Results

6.1 Data Cleaning

After data inspection, no sessions were excluded from the final analysis.

6.2 Behavioral Descriptives

The distribution of response time and mean accuracy is visualized in Figure 10. The response time across individuals had a weighted mean of 2.94s (*SD* = 0.68), whereas the mean accuracy was 0.88 (*SD* = 0.08). An overview of the summary statistics of response times, accuracy, median number of trials, median number of observed facts, and the median number of repetitions per fact are displayed in Table 5 below.

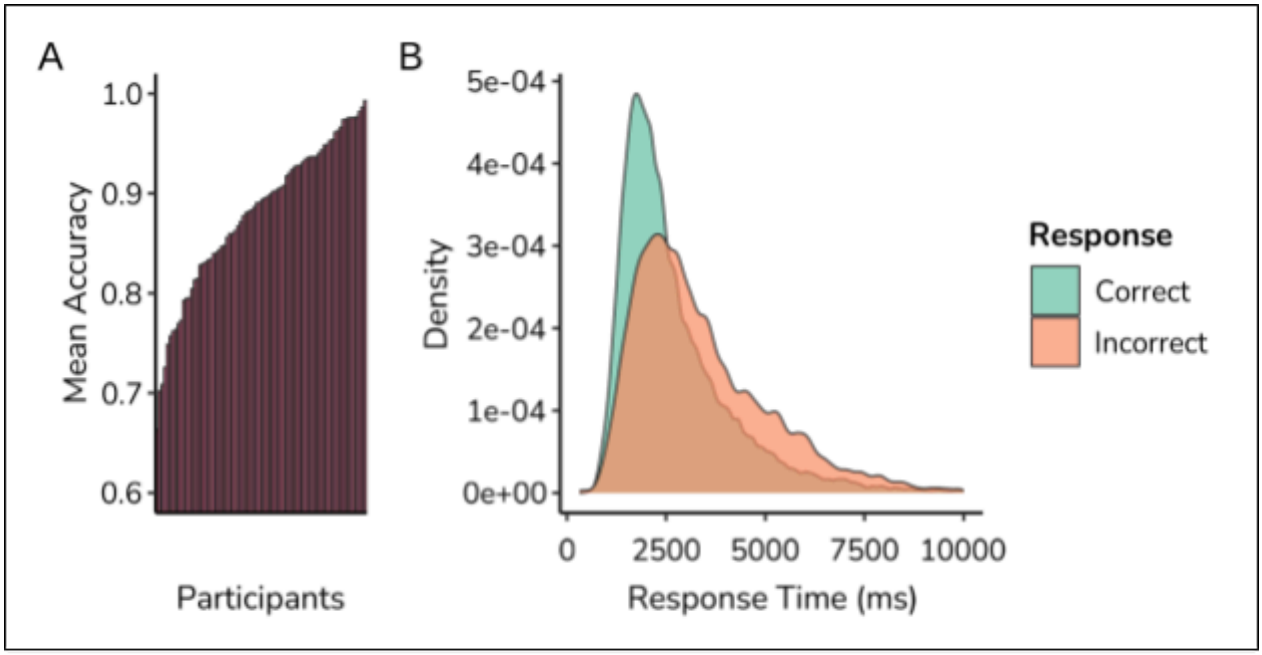

Figure 10. (A) Distribution of mean accuracy across participants. (B) Distribution of correct and incorrect response times.

Table 5: Behavioral Descriptives

Manipulation	Response Time (s)	Accuracy	Median trials	Median observed facts	Median number of repetitions
Control Condition	2.90 (0.64)	0.88 (0.08)	139.5	19	6

Table 5: Behavioral Descriptives

Fast-Latency Condition	2.97 (0.69)	0.88 (0.08)	136.5	19	6
Slow-Latency Condition	2.95 (0.72)	0.88 (0.10)	136.0	20	6

Behavioral descriptives including weighted-average response time in seconds, weighted-average accuracy, median trials per session, median observed facts, median number of repetitions.

6.3 Hypothesis Testing

6.3.1 *Confirmation Prompt.* An overview of the distribution of errors across the conditions can be found in Table 6 below. It can be observed that the proportions of error types remained relatively constant across conditions, $X2$ (4) = 3.39, p = .495. A total of 835 incorrect first attempts were prompted, whereas 862 correct first attempts were prompted, confirming that we were successful in balancing prompts after errors and prompts after correct answers. A visualization of when trials were prompted over time for three participants is shown in Figure S2 in the supplements. Figure 11 displays the relationship between error latency and item difficulty, here it can be observed that, just as in the retrospective data analysis, the proportion of fast-latency errors increases as item difficulty decreases.

Table 6: Error Distribution

Manipulation	Fast-Latency Error	Mid-Latency Error	Slow-Latency Error
Control Condition	0.32 (0.16)	0.36 (0.15)	0.32 (0.24)
Fast-Latency Condition	0.32 (0.17)	0.35 (0.16)	0.33 (0.24)
Slow-Latency Condition	0.34 (0.20)	0.33 (0.14)	0.33 (0.24)

Distribution of error types. Entries show the mean proportion of error types within each manipulation, with standard deviations across participants shown in parentheses. Means are computed at the participant level and are not weighted by the number of trials.

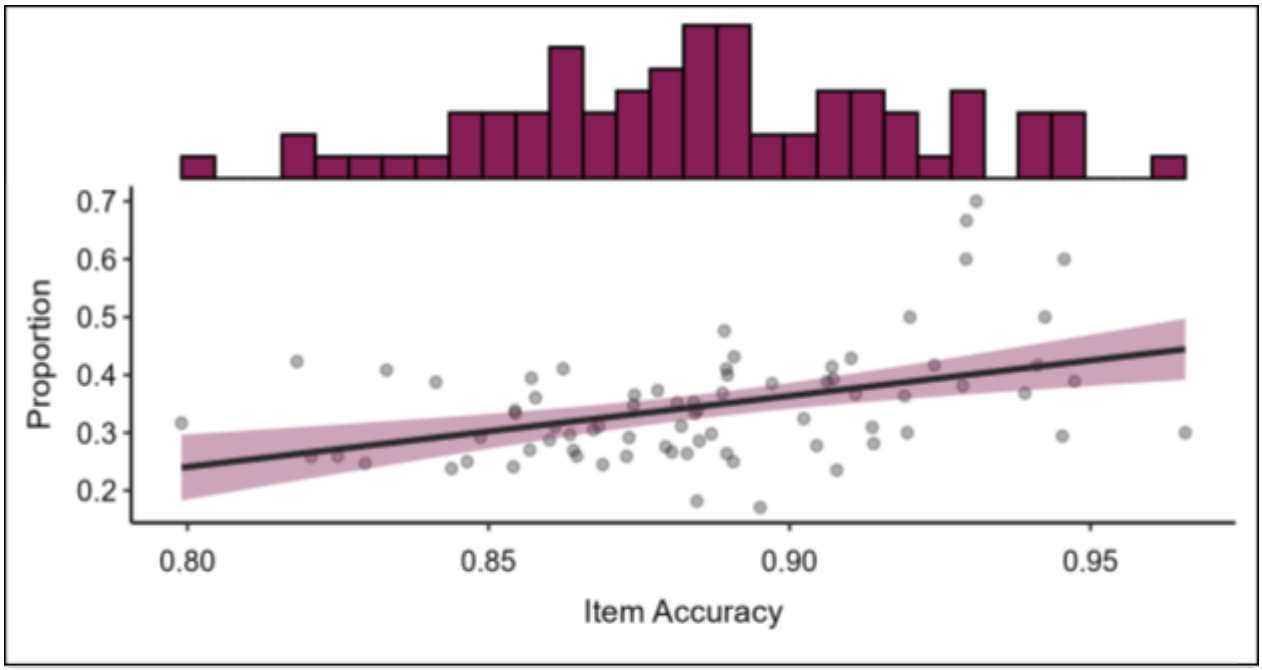

Figure 11. Proportion of fast-latency errors as a function of item accuracy. The marginal distribution of item accuracy is displayed above.

An overview of the correction patterns after prompting for confirmation on initially correct and incorrect trials can be found in Table 7 below. The table shows that, when the confirmation prompt followed a correct response, participants almost never changed their answer (2.1%), but when the prompt followed an error, they changed their answer on 26.2% of trials.

Table 7: Correction Patterns

Error Type	Initially Correct		Initially Incorrect		
	Still correct	Erroneously corrected	Accurately corrected	Still incorrect, same answer	Still incorrect, different answer
Fast-Latency	443 (99.5%)	2 (0.5%)	131 (33.2%)	248 (62.8%)	16 (4.0%)
Slow-Latency	401 (96.2%)	16 (3.8%)	54 (12.3%)	368 (83.6%)	18 (4.1%)

Correction patterns after a confirmation prompt. Percentages show how participants' second responses differed depending on whether the first attempt was correct or incorrect, for fast- and slow-latency errors.

A GLMER was performed to investigate whether there were differences in correction rate between fast- and slow-latency errors, of which the ANOVA table can be found in Table 8 below, the fixed-effect estimates can be found in Table S2 in the Supplements. No apparent patterns indicating violations of model assumptions were found (see Figure S3 in the Supplements). No significant effects on final accuracy were observed for the stimulus set and block number control variables. An overall effect of the latency type (fast-latency vs. slow-latency) on final accuracy was found, as well as an overall effect of initial correctness (accuracy of the initial response). No significant difference was found for the interaction effect of latency type and initial correctness. In concrete terms, this means that fast and slow (error) correction rates were similar regardless of participants' initial correctness; the initial correctness did not moderate the relationship between latency type and correction. Table 9 summarizes differences in final accuracy between fast- and slow-latency response by initial correctness. A visualization of the marginal means of the interaction effect can be found in Figure 12. The data suggests that fast-latency errors are corrected significantly more often than slow-latency errors. Given that the prompt itself is uninformative about accuracy, the slow-latency error correction rate was just below the guess rate of 25% . Moreover, participants were significantly more likely to stick with the correct answer when their initial response was fast than when it was slow.

Table 8: Correction Rates ANOVA Table

Predictor	X^2	*DF*	*p*
Stimuli Set	1.69	2	.429
Block	2.47	2	.291
Latency	37.17	1	< .001*
Initial Correctness	324.67	1	< .001*
Latency*Correctness	0.83	1	.361

*Analysis of Deviance (Type II, Wald) table of the GLME of accuracy on the second attempt as a function of stimulus set, block, response latency type, accuracy on the first attempt, and the interaction of response latency type * first attempt accuracy.*

Table 9: Pairwise Comparisons

First Attempt	Difference	95% CI		*Z*	*p*
	Fast - Slow	LB	UB		
Correct	0.03	0.01	0.05	2.86	.004*
Incorrect	0.22	0.15	0.28	6.21	< .001*

Difference in accuracy on the second attempt in proportion points between fast and slow, for correct and incorrect first attempts. p-values are uncorrected.

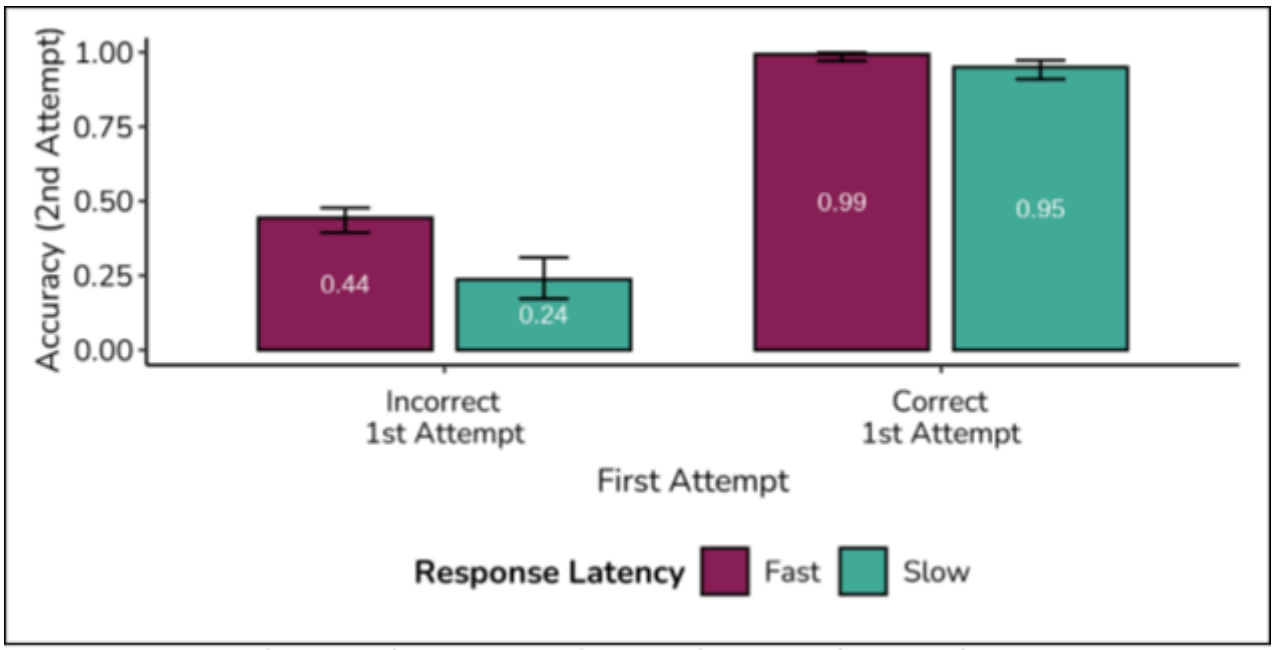

Figure 12. Marginal means of accuracy on the second attempt after a confirmation prompt.

6.3.1.1 *Confirmation Prompt: Learning Effect.* Finally, we wanted to see if effects of successful self-correction carry over to subsequent trials, thus we explored whether the follow-up accuracy at the subsequent trial of the accurately corrected trial differed from trials that were not accurately corrected after prompting for revision. We performed a GLMER on the initially incorrect trials where a prompt occurred, with the block number and stimulus set as control variables, and the error type at lag 1 and whether the trial was correctly corrected at lag 1 plus their interaction as predictors. Here, the reference category was 'Fast - Correctly Corrected'. The ANOVA table can be found in Table 10 below, whereas the fixed-effect estimates can be found in Table S3 in the supplements. No apparent patterns indicating violations of model assumptions

were found (see Figure S4 in the Supplements). Although the ANOVA table indicates no significant or interaction effect was found, the fixed-effect slope of 'Accurately Corrected' differed marginally significant from 0. Therefore, we conducted a post-hoc analysis of the interaction effect, of which a visualization of the marginal means is displayed in Figure 13. The contrast analysis (Table 11) suggests that fast errors that are accurately revised after prompting are answered with higher accuracy as compared to fast errors that aren't accurately revised. This respective pattern was not found for slow-latency errors. Moreover, a marginally significant difference between fast trials that were correctly corrected versus fast trials in the control condition (not prompted, nor corrected) was observed (p = .050) . Although on average the accurately corrected fast errors did have a higher accuracy at a subsequent trial of the same item compared to unsuccessfully corrected fast errors (estimated difference: 11 percentage points; p = .015), the difference between *accurately corrected, fast-latency* errors compared to *accurately corrected, slow-latency* errors was not significant.

Table 10: Learning Effect (n+1) ANOVA Table

Predictor	X^2	*DF*	*p*
Stimulus Set	7.38	2	.025*
Block	8.50	2	.014*
Correction	1.86	1	.173
Error Type	0.06	1	.801
Correction*Error Type	2.14	1	.144

*Analysis of Deviance (Type II, Wald) table of the GLME of accuracy on the subsequent trial of the same item (n+1) as a function of stimulus set, block, error latency type on the first attempt,, accuracy (on the second attempt for incorrect first attempts, and on the first attempt of control trials) , and the interaction of error latency type * (second attempt) accuracy.*

Table 11: Pairwise Comparisons of Correction*Error Type (n+1)

Comparison	Difference	95% CI		*Z*	*p*
		LB	UB		

Table 11: Pairwise Comparisons of Correction*Error Type (n+1)

By correction					
Acc. - inacc. (fast)	0.11	0.02	0.21	2.42	.015*
Acc. - control (fast)	0.08	0.00	0.17	1.96	.050
Inacc. - control (fast)	-0.03	-0.10	0.04	-0.81	.416
Acc. - inacc. (slow)	-0.02	0.00	0.15	0.06	.956
Acc. - control (slow)	-0.03	-0.17	0.11	-0.41	.684
Inacc. - control (slow)	-0.03	-0.10	0.03	-1.02	.309
By latency					
Fast - Slow (acc).	0.10	-0.05	0.26	1.34	.181
Fast - Slow (inacc.)	-0.01	-0.08	0.07	-0.18	.860
Fast - Slow (control)	-0.01	-0.07	0.05	-0.30	.760

Differences in accuracy on the subsequent trial of the same item (n+1), expressed in proportion points, depending on how first-attempt errors were corrected. Comparisons are shown separately by second-attempt outcome (accurate, inaccurate, control) and by latency (fast vs. slow). p-values are uncorrected. Abbreviations: Acc. = Accurately Corrected; Inacc. = Still Incorrect.

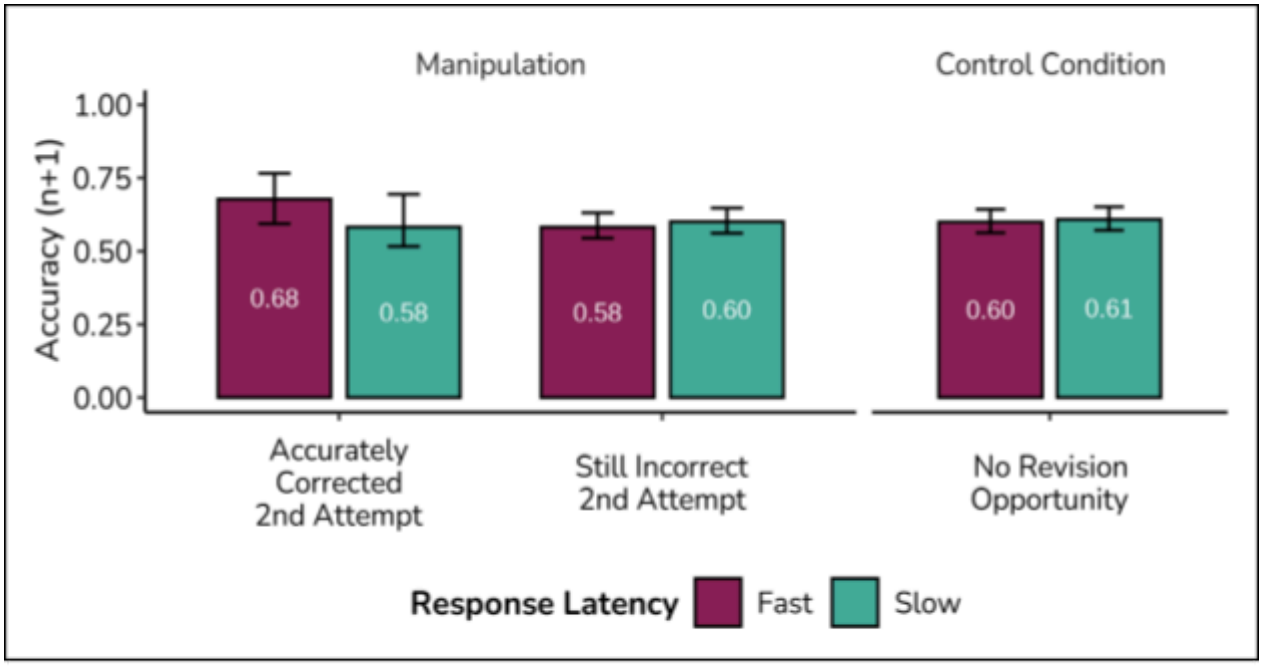

Figure 13. Accuracy on the subsequent trial of the same item after an incorrect-prompt trial, with the non-prompted trials from the control condition on the right.

6.3.2 *Effect of Feedback on Accuracy* A GLMER was performed to investigate whether implementing confirmation prompt feedback improves learning, of which the ANOVA table can be found in Table 12 below, the fixed-effect estimates can be found in Table S4 in the Supplements. No apparent patterns indicating violations of model assumptions were found (see Figure S5 in the Supplements). A significant effect on accuracy was found for the block number (likely related to a learning effect of the task) and the stimulus set control variables. Accuracy improved significantly only for the block where fast-latency responses were prompted (see Table 13).

Table 12: Final Accuracy ANOVA Table

Predictor	X^2	*DF*	*p*
Stimuli Set	8.75	2	0.02*
Block	37.80	2	< .001*
Manipulation	18.01	2	< .001*

Analysis of Deviance (Type II, Wald) table of the GLME of final accuracy (first response for non-prompted trials, and second response for prompted trials) as a function of stimulus set, block, and manipulation type.

Table 13: Pairwise Comparisons of Manipulation

Final Accuracy	Difference	95% CI		*Z*	*p*
		LB	UB		
Slow - Control	0.00	0.00	0.01	0.95	.341
Fast - Control	0.02	0.01	0.03	3.89	< .001*
Fast - Slow	0.01	0.01	0.02	3.01	.003*

Differences in final accuracy between the manipulations, expressed in proportion points.

p-values are uncorrected.

7 Overall Discussion

In this study, we examined whether the speed of erroneous responses in a spaced-repetition recognition memory task predicts future performance. Our analysis of previously collected spaced-repetition retrieval practice with multiple-choice questions data revealed that fast errors result in higher accuracy on subsequent trials of the same item than slow errors, both in an implicitly motivated group (TransplantLines) and an extrinsically motivated group (Klett & Balmer). Building on these findings, we conducted an empirical study to assess whether fast errors provide a more suitable opportunity for self-correction than slow errors, by prompting participants to reconsider these fast and slow responses. Our results revealed that, indeed, fast errors are accurately revised more often than slow errors.

7.1 Experiment

The results from the retrospective analysis motivated us to investigate whether the distinction between fast and slow errors was also observable in how these errors are monitored during a learning task, and whether this could be leveraged to provide more tailored error feedback based on how an error occurred. More specifically, since fast errors are often detected and corrected internally (Novikov et al., 2017) as opposed to slow errors, feedback could encourage reflection rather than providing the correct answer directly for these fast errors. To test this, we set up an experiment where the feedback was tailored to the error type. The manipulation involved a confirmation-prompt feedback mechanism that was implemented in two distinct experimental blocks: one targeting fast-latency errors and the other targeting slow-latency errors. The prompt offered participants the opportunity to confirm or revise their initial response. To maintain experimental ambiguity and prevent participants from associating the confirmation prompt with an error cue, it was also introduced during a subset of correct trials with a latency corresponding to the block type. Following Novikov and colleagues (2017), we hypothesized that fast-latency errors would be corrected significantly more often than slow-latency errors.

Our results support this hypothesis, showing that in 44% of fast-latency errors individuals would accurately revise their answer after prompting for confirmation, versus 24% of the prompted slow-latency errors (Figure 12): a significant difference. In addition, correct responses given with fast latencies were changed to incorrect responses far less frequently than correct responses given with slow latencies, reinforcing the interpretation that slow-latency responses reflect heightened uncertainty, also observed in visual search tasks (Hodsoll & Humphreys, 2001) and inhibitory control tasks (Sabri et al., 2001). Notably, only 2 out of 445 of initial correct responses that were followed by a confirmation prompt were changed to an incorrect answer, implying that prompting after fast-latency errors has little negative side-effects on learning.

Additionally, we find limited evidence for the hypothesis that confirmation prompting supports learning. The results suggest that accurately correcting the incorrect first attempt had a positive effect on learning, but only for fast-latency errors. Fast-latency accurately-corrected first attempts had a significantly higher accuracy on the subsequent trial of the same item compared to fast-latency incorrect first attempts that were not accurately corrected (68% vs. 58%), as well as a marginally-significant higher accuracy compared to fast-latency errors during in the control condition (68% vs 60%). A successful correction following a confirmation prompt alone was not enough to boost subsequent performance: accuracy following a successfully corrected

slow-latency error was no different from accuracy following an unrevised or unsuccessfully corrected error. Notably, we did not find the expected difference between fast-latency and slow-latency in accuracy at the subsequent trial of the same item errors in the control condition (60% vs. 61%), given the observed difference in the retrospective data analysis. This suggests that the learners in our sample learned somewhat differently than the learners from the TransplantLines and Klett & Balmer datasets.

On the block level, the confirmation prompt resulted in significantly higher final accuracy in the block where fast-latency responses were prompted, compared to both the control (two percentage points difference) and the slow-latency condition (one percentage point difference). The slow-latency condition did not differ significantly from the control, likely because a larger proportion of correct first-attempt responses were changed to incorrect second attempts in this condition. Taken together, these results suggest that self-correction is most beneficial when learners are able to correct fast errors, which are more likely to reflect familiarity-based or impulsive responding rather than genuine retrieval failure. By contrast, slow errors are likely harder to correct because they often reflect genuine retrieval failure, limiting any potential benefit from self-correction.

7.2 Strengths and Limitations

First, our methodology combined an exploratory analysis of existing data with a targeted experimental manipulation. This approach allowed us to ground our intervention in naturally occurring behavioral patterns observed over a large dataset while simultaneously testing causal hypotheses under controlled conditions. This contrasts with studies relying solely on either purely observational or purely experimental designs. Second, we introduced a novel and behaviorally meaningful implementation of error classification based on response-time deviations using a model-based approach informed by the ACT-R cognitive architecture, allowing us to distinguish between likely impulsive (fast) and uncertainty-/attention-related (slower) errors. A key advantage of this model-based classification is its ability to classify errors at the level of individual trials, making it immediately applicable in a real-time and online setting for adaptive interventions. The approach performs a ‘context-aware’ classification of error speed by incorporating the activation of the memory chunk, offering a more nuanced metric for fast errors than a simple fixed-time threshold. However, our current classification of slow errors is functionally similar to a fixed-response time decision rule, suggesting future refinement could

involve estimating a dynamic threshold for each learner (e.g., using a data-driven approach like fitting a Drift-Diffusion Model to the distribution of responses). Furthermore, the precise mapping between these ACT-R classifications and distinct underlying cognitive processes remains an open question, and incorporating retrospective introspective judgments from participants could be a valuable part of the model validation process. Third, the study's generalizability is enhanced by its diverse participant pools and stimulus modalities. Our analyses drew on data from two distinct groups: one implicitly motivated group (TransplantLines), who volunteered without the promise of external recompense such as monetary payment or course credit, indicating a self-directed interest in the research subject or an altruistic desire to contribute to science; and another explicitly motivated group participating as part of homework assignment (Klett & Balmer). This distinction is critical because the presence of an external incentive can alter the cognitive approach and investment of the participants (Albrecht et al., 2014). Furthermore, the design was robust across stimulus types, with image-based stimuli for the TransplantLines sample and word-based stimuli for the Klett & Balmer sample. This broadens the applicability of our findings beyond a single demographic or task format. Fourth, the intervention itself was designed for practical feasibility and scalability. By tailoring feedback to specific error types, we developed a flexible intervention (the confirmation prompt) that is both low-cost to implement and easily scalable within adaptive learning technologies. This represents a significant design strength for real-world application.

Nonetheless, the study also has important limitations. First, to ensure that response caution was sufficiently low so that individuals would not optimally consider each answer option to ensure enough fast-latency errors (i.e., enough power), we instructed participants in the experiment to “Try to respond quickly, making errors is fine as you learn”. As a result, the experimental data might not fully reflect how participants would behave in a typical learning environment. Moreover, this instruction likely interacted with individual differences in baseline response caution and in sensitivity to making errors, meaning that the manipulation may not have affected all learners in the same way. Still, accuracy was high with an average accuracy of 0.88, with the lowest performance being 0.66. Moreover, the instruction resulted in an even distribution of fast-, mid- and slow-latency errors across the three conditions. Second, although most of the stimuli were randomly selected from the most difficult ranked outlines from Wilschut et al., (2023), after initial testing we had to replace some of the stimuli as they were deemed

uninformative for the task (i.e., they had an accuracy of ~1.0), likely because the outlines were too distinctive, potentially rendering the stimulus selection not completely unbiased. Third, our sample consisted of exclusively first-year psychology students, meaning that generalization is limited. While this group is a plausible user base for adaptive learning tools like the MemoryLab system, additional research is required to assess whether the findings generalize beyond this population. Lastly, our methodology prompted responses even for correct trials to reduce participants' association of the prompt exclusively with mistakes. While effective in this regard, this approach may have influenced participants' subjective experience of the task and their response strategies in ways not directly measurable, independent of any effects on accuracy Such experiential effects may be particularly relevant for long-term outcomes such as engagement or perseverance. Finally, although successful correction was associated with higher accuracy on the subsequent trial of the same item, it remains unclear whether this effect reflects the initial error not being a genuine retrieval failure, or whether self-correction improved short-term learning that may subsequently generalize to long-term learning. Future work should include a delayed posttest to examine whether these short-term accuracy benefits persist over time.

7.3 Implications

The findings from this study offer several important implications for both cognitive research and applied learning contexts. First, our results highlight the value of using response time as an informative behavioral signal when examining errors in spaced-repetition retrieval practice with multiple-choice questions. Rather than treating all mistakes as equivalent, distinguishing fast from slower errors provides a useful descriptive distinction: fast errors tend to be more readily corrected than slow errors, and correction of fast errors can produce learning benefits. This distinction offers a nuanced way to characterize variability in error patterns without assuming a single, uniform source of mistakes. This differentiation provides a more nuanced understanding of error mechanisms, which can inform and refine cognitive models of memory retrieval and the learning process itself.

Second, these results have direct implications for learner modeling. Careless or fast-latency errors can lead learner models to make incorrect inferences about a learner's underlying knowledge state unless such models explicitly incorporate mechanisms to detect or accommodate these errors. In many adaptive learning frameworks, such as Bayesian Knowledge Tracing models (Corbett & Anderson, 1994), this issue is addressed via parameters that capture slips or

lapses, *p*(slip). In the present work, we demonstrate an activation-based solution within ACT-R, showing how response latency can be used as an indicator for errors that reflect transient lapses and those that reflect genuine knowledge gaps. When learner models fail to account for such distinctions, knowledge levels may be systematically underestimated, potentially resulting in inefficient sequencing of practice, unnecessary repetition, and a degraded learning experience.

Third, the observed effectiveness of the confirmation prompt for fast-latency errors demonstrates that subtle, behaviorally informed interventions can significantly improve immediate performance. This confirmation prompt not only helped participants recover from fast errors but also improved the likelihood of future correct responses for the same item. This suggests that learners can genuinely benefit from momentary opportunities to reflect and revise, especially in cases where errors stem from possible hastiness rather than deeper misunderstanding. This approach has practical relevance for adaptive learning technologies and individualized education strategies. Systems that monitor response speed could implement dynamic, real-time feedback like the confirmation prompt to encourage intentional revisions which are a key component of effective self-regulated learning and a process known to deepen understanding (Winne & Hadwin, 1998). Moreover, given that prior research has linked errors to increased dropout rates in learning tasks (Drake et al., in press), integrating such prompts could serve as a low-cost intervention to stabilize engagement and enhance retention in educational or training contexts, particularly for individuals who are prone to impulsive or fast errors, such as those with developmental disorders like ADHD. For learners with ADHD, who often exhibit impulsivity and difficulties with error monitoring (e.g., Barkley, 1997), confirmation prompts offer a structured opportunity for metacognitive reflection and self-correction, potentially mitigating the negative impact of rapid, uncorrected responses on learning and motivation.

8. Conclusion

This study underscores the importance of response speed in understanding the nature and consequences of spaced-repetition fact learning errors. Through both retrospective and experimental approaches, we found that error-response latency signals differences in the processes underlying recognition-memory errors. Distinguishing between fast and slow errors - paralleling distinctions made in decision-making research, could optimize learning. Consistent with work showing that fast errors are often internally detected, we found that fast errors provide an opportunity for effective self-correction: a brief confirmation prompt improved both

immediate recovery and later accuracy, suggesting that quick revision opportunities can support learning when mistakes stem from hastiness rather than deeper misunderstanding. Adaptive learning systems that monitor response speed could deliver real-time prompts to encourage intentional revision. Overall, our results underscore the potential of erroneous response time as a meaningful behavioral marker for both understanding learning processes and designing subtle, effective interventions.

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Supplementary Material

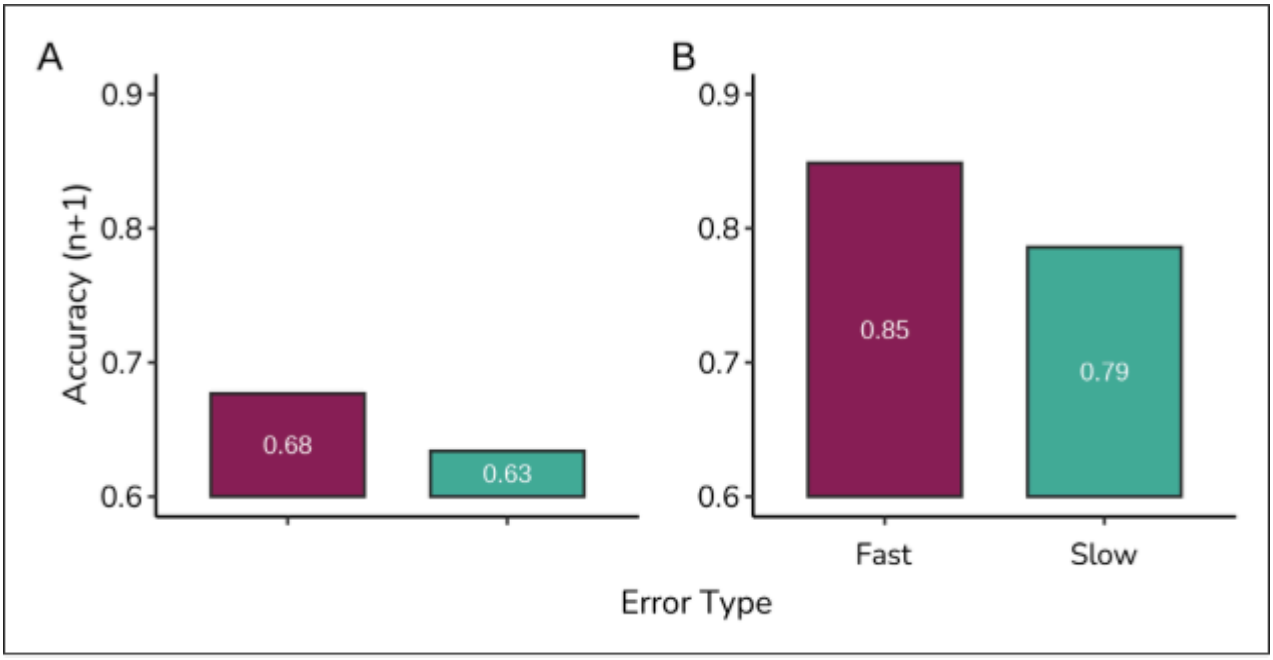

Figure S1. Accuracy at the subsequent trial by error type for all participants when slow-latency errors are quantified as errors slower than 3.8s (1.5 times the forgetting threshold)

Table S1: Modifications to Stimuli	
Original	Modified

Table S1: Modifications to Stimuli	

Overview of all stimulus modifications.

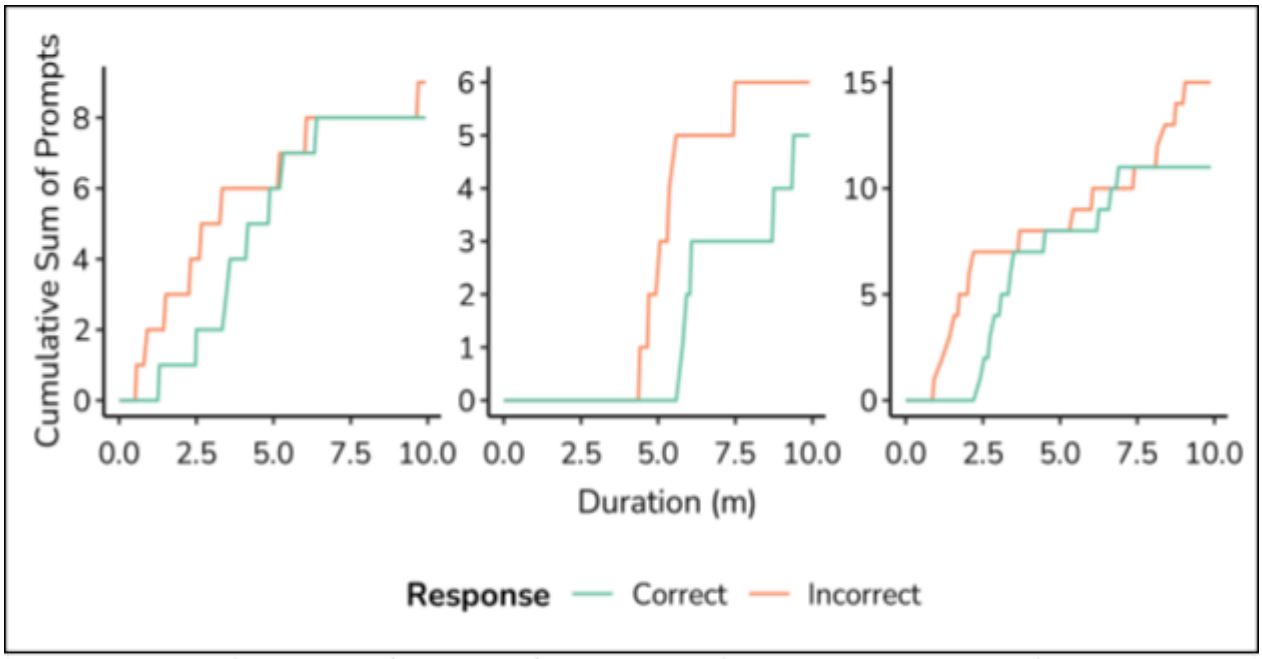

Figure S2. Cumulative sum of prompts after correct and incorrect responses in the Fast-Latency condition.

Table S2: Correction Rates Fixed Effects Estimates

Predictor	Estimate [b]	95% CI		*z*	*p*
		LB	UB		
Intercept [a]	3.69	-	-	-	-
Stimulus Set					
Asia	0.25	-0.39	0.88	0.76	.445
Europa/Latin-America	-0.19	-0.81	0.43	0.60	.549
Block					
2	0.15	-0.35	0.65	0.59	.553
3	0.44	-0.11	0.99	1.57	.117
Response Type					
Fast	2.01	0.51	3.51	2.62	.009
First Attempt					
Incorrect	-5.99	-6.7	-5.27	-16.48	< .001
Response Type * First Attempt					
Fast*Incorrect	-0.72	-2.27	0.83	0.91	.361

*Mixed-effects logistic regression predicting accuracy on the second attempt as a function of stimulus set, block, response latency type, accuracy on the first attempt, and the interaction of response latency type * first attempt accuracy.*

[a] *Intercept = log-odds of being accurate in Block 1 on the Africa stimulus set , given a correct first attempt and slow response.*

[b] *Log-odds*

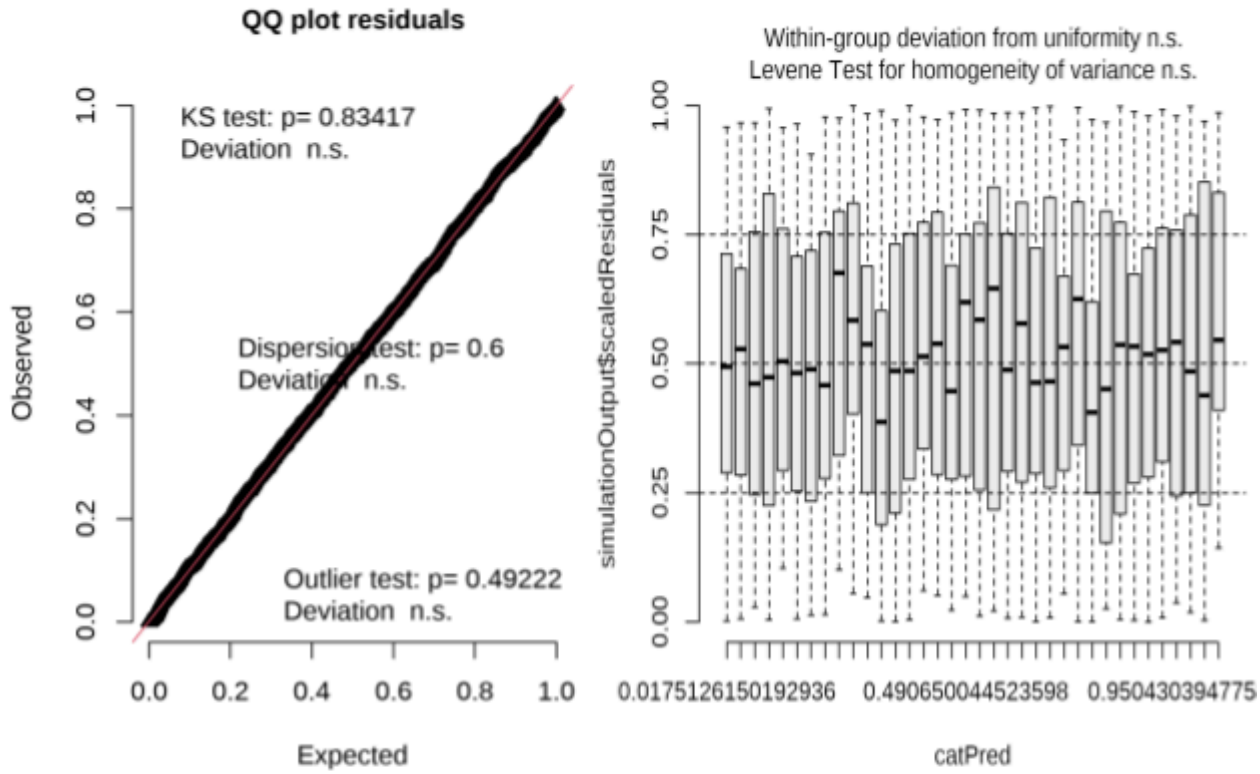

Figure S3. Residual diagnostics for the generalized linear mixed-effects model predicting accuracy on the second attempt after prompting for confirmation. Simulated residuals were obtained using the DHARMa package (Hartig, 2016). (Left) Quantile-Quantile (Q-Q) plot of standardized residuals, showing conformity to expected distribution. (Right) Residuals plotted against predicted values, with no apparent patterns indicating violations of model assumptions.

Table S3: Accuracy (N+1) Fixed Effects Estimates

Predictor	Estimate [b]	95% CI		*z*	*p*
		LB	UB		
Intercept [a]	0.88	-	-	-	-
Stimulus Set					
Asia	0.25	-0.05	0.54	1.62	.106
Europa/Latin-America	-0.07	-0.35	0.22	-0.46	.642
Block					
2	0.21	-0.07	0.49	1.45	.146
3	0.35	0.05	0.65	2.25	.024*
Error Type					
Slow	0.06	-0.28	0.40	0.34	.735
Second Attempt					
Accurately Corrected	0.47	-0.07	1.01	1.69	.091
Still Incorrect	-0.12	-0.48	0.25	-0.63	.526
Response Type * Second Attempt					
Fast*Accurately Corrected	-0.64	-1.54	0.25	-1.41	.158
Fast * Still Incorrect	0.07	-0.43	0.56	0.27	.790

*Mixed-effects logistic regression predicting accuracy on the subsequent trial of the same item (n+1) as a function of stimulus set, block, error latency type on the first attempt, accuracy (on the second attempt for incorrect first attempts, and one the first attempt of control trials) , and the interaction of error latency type * (second attempt) accuracy.*

[a] *Intercept = log-odds of being accurate in Block 1 on the Africa stimulus set , given a correct first attempt and slow response.*

[b] *Log-odds*

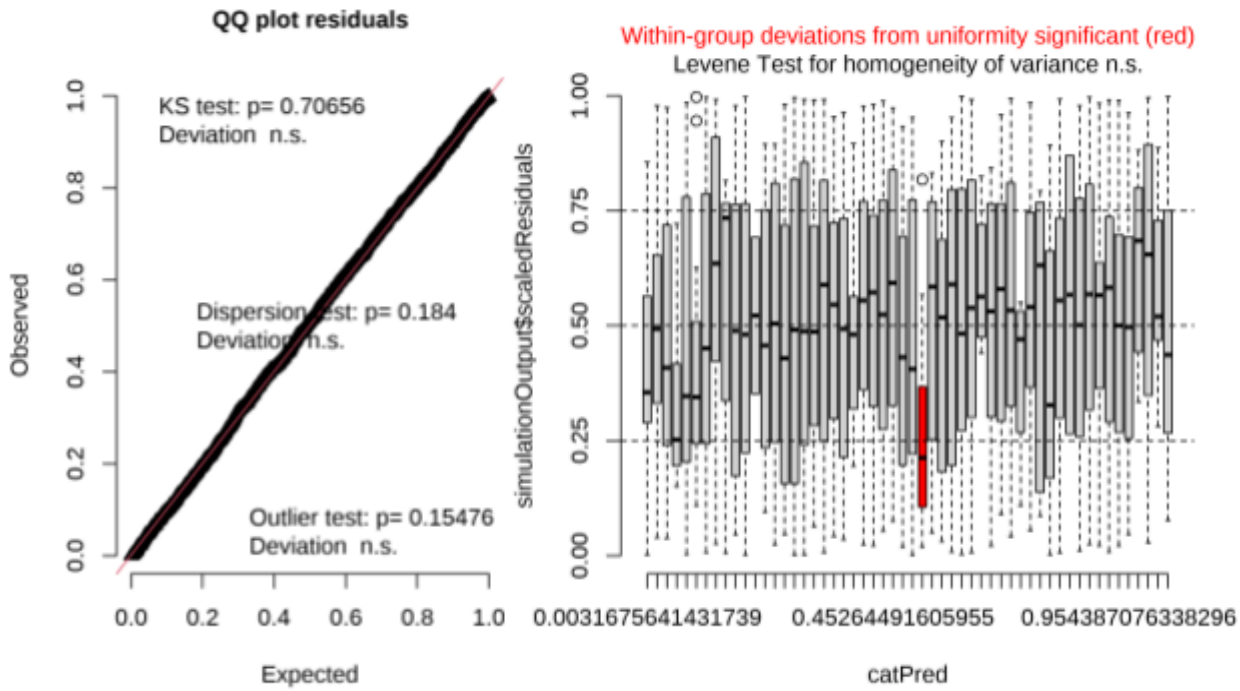

Figure S4. Residual diagnostics for the generalized linear mixed-effects model predicting n+1 accuracy after a fast/slow error on the first attempt. Simulated residuals were obtained using the DHARMa package (Hartig, 2016). (Left) Quantile-Quantile (Q-Q) plot of standardized residuals, showing conformity to expected distribution. (Right) The plot of residuals against predicted values revealed some statistically significant deviations from uniformity for certain combinations of factor levels; however, a visual inspection suggested these departures were not substantial enough to compromise the model fit.

Table S4: Final Accuracy Fixed Effects Estimates

Predictor	Estimate [b]	95% CI		*z*	*p*
		LB	UB		
Intercept [a]	1.83	-	-	-	-
Stimulus Set					
Asia	0.25	0.07	0.43	2.74	.006
Europa/Latin-America	0.04	-0.14	0.21	0.39	.70
Block					
2	0.23	0.14	0.32	5.08	< .001
3	0.25	0.16	0.34	5.51	< .001
Manipulation					
Slow-Latency	0.07	-0.02	0.15	1.55	.120
Fast-Latency	0.19	0.10	0.28	4.21	< .001

Mixed-effects logistic regression predicting final accuracy (first response for non-prompted trials, and second response for prompted trials) as a function of stimulus set, block, and manipulation type

[a] *Intercept = log-odds of being accurate in Block 1 on the Africa stimulus set, during the control condition.*

[b] *Log-odds*

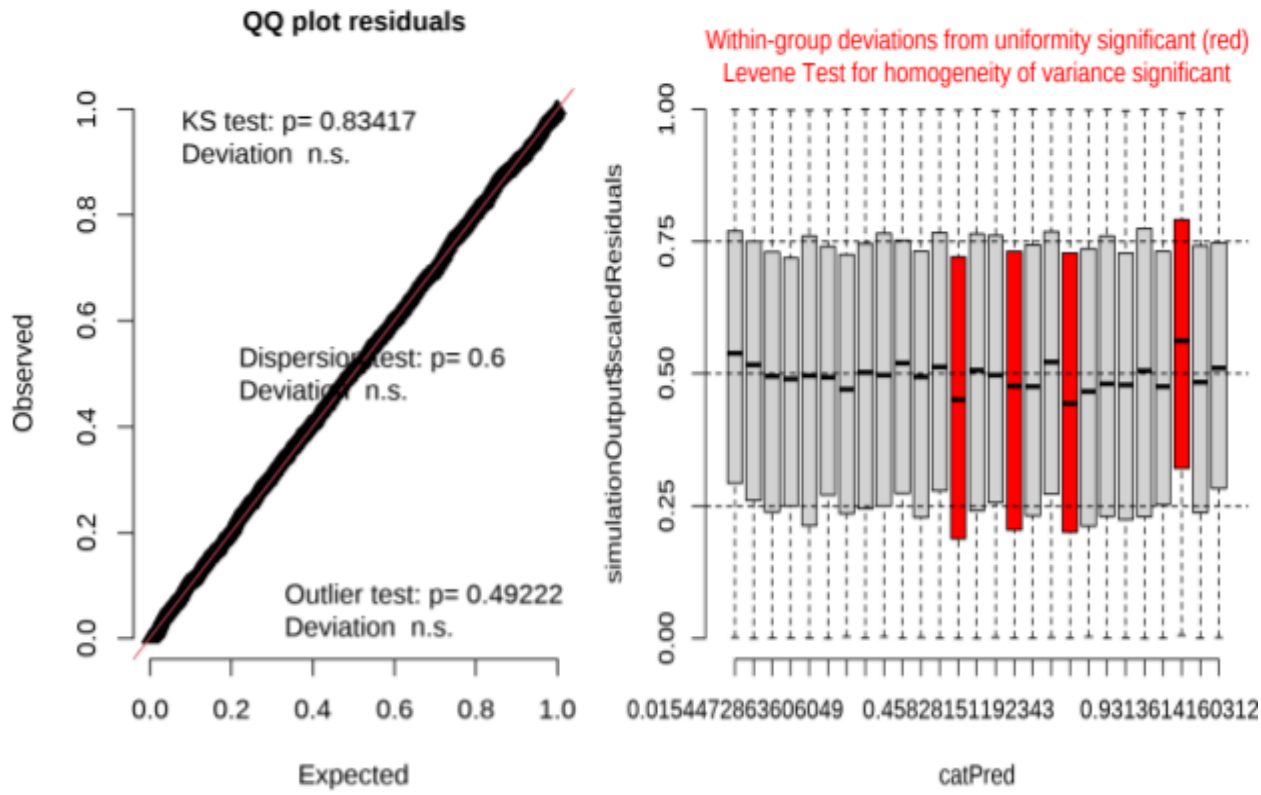

Figure S5. Residual diagnostics for the generalized linear mixed-effects model predicting final accuracy across all trials. Simulated residuals were obtained using the DHARMa package (Hartig, 2016). (Left) Quantile-Quantile (Q-Q) plot of standardized residuals, showing conformity to expected distribution. (Right) The plot of residuals against predicted values revealed some statistically significant deviations from uniformity for certain combinations of factor levels; however, a visual inspection suggested these departures were not substantial enough to compromise the model fit.